Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

Name	Type of File	Size	Page
EcolLett_RESPONSE.pdf	Author Response	206.9 KB	Page 3
EcolLett_R1_SI.pdf	Supplementary Material for Review	5.4 MB	Page 19
EcolLett_R1.pdf	Main Document - LaTeX PDF	3.9 MB	Page 48
Supplementary_LaTeX.zip	LaTeX Supplementary File	4.8 MB	*Not converted to PDF*
Combined Submission Document.pdf	Combined Submission Document (Novelty Statement, Cover Letter, Title Page)	102.5 KB	Page 75

ECOLOGY LETTERS

Letter

Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

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Authors

Dr. Jacob K. Moutouama
Corresponding Author
Submitting Author

ORCiD
https://orcid.org/0000-0003-1599-1671

Affiliations
- Department of Botany, University of British Columbia, Vancouver, BC, Canada
- African Centre for DNA Barcoding, University of Johannesburg, Johannesburg, South Africa

Dr. Joshua C. Fowler

Affiliations
- Department of Environmental Studies, University of Colorado Boulder, Boulder, CO, USA

Mr. Akiem M. Gough

ORCiD
https://orcid.org/0009-0003-2604-4773

Affiliations
- Department of BioSciences, Rice University, Houston, TX, USA

Mrs. Julia Martin

Affiliations
- Department of BioSciences, Rice University, Houston, TX, USA

Mr. Christopher C. Oxley

Affiliations
- Department of BioSciences, Rice University, Houston, TX, USA

Mr. Karl Schrader

Affiliations

- Department of BioSciences, Rice University, Houston, TX, USA

Mr. Karis Williams

Affiliations

- Department of BioSciences, Rice University, Houston, TX, USA

Prof. Tom E.X. Miller

Affiliations

- Department of BioSciences, Rice University, Houston, TX, USA

Manuscript 7613110 – Response to reviewers

To the editorial board,

Thank you for the opportunity to submit a revision of our manuscript, "Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses," for your consideration. Our major changes include the following:

1. We revised our conceptual framing to incorporate more recent extensions of the stress-gradient hypothesis (SGH), which recognize that positive interactions can weaken or reverse under sufficiently severe or compounding stress, rather than increasing monotonically with stress as in the classic SGH. This revision spans the Introduction (hypotheses) and Discussion, and clarifies that our finding of declining endophyte benefits with increasing aridity is consistent with this more nuanced framework rather than a simple contradiction of SGH (Reviewers 1 & 2).

2. We conducted a new sensitivity analysis excluding the two field sites (KER and SON) affected by disturbances followed by replanting events, to directly test whether this disturbance or the associated variability in herbivore-exclusion efficacy influenced our results. Posterior estimates of endophyte effects were nearly identical with and without these sites across all species, vital rates, and treatments, and we have added this analysis, along with a new supplementary figure and table, to the manuscript (Reviewer 1).

3. We expanded the Discussion to address the broader biological implications of our results, including the potential consequences of climate-driven shifts in symbiosis costs and benefits for endophyte prevalence, host-range dynamics, and symbiont retention, as well as the role of variation in endophyte alkaloid chemistry in explaining species-specific herbivory responses (Reviewers 1 & 3). We also corrected a data-interpretation error in the reported herbivory results for *A. hyemalis* (Reviewers 1 & 3).

We describe these and other changes in greater detail below, where we reproduce comments from the reviewers and provide our point-by-point responses. All of our changes are denoted in the manuscript with Mahogany font. We think the review process has strengthened our manuscript and hope you agree that it is now suitable for publication.

On behalf of my coauthors and myself,
Jacob Moutouama

Response to Reviewer 1

Comment 1: *"The authors test how the benefits of an endophyte symbiosis interacted with climatic stress (precipitation) gradient. They found that the benefits were largely lost with increasing aridity. This is a very strong study that was well designed and the analyses were well interpreted. The results are not entirely novel due to the well documented variation in endophyte benefits depending on the host and endophyte species. Even though the results are not novel, the study is one of the most complete analyses of the relationship between climate and endophyte effects on vital rates under field conditions. I believe this study is a solid contribution to the literature on plant microbial symbiosis. It is also a very well written manuscript."*

Response: We thank Reviewer 1 for these positive and constructive comments. We appreciate the recognition of the study's experimental design, analytical approach, and contribution to the literature on plant–microbial symbiosis. We agree that variation in the benefits of endophyte symbiosis among host and endophyte species has been well documented. We believe that our study extends this literature by explicitly testing how climatic stress modifies endophyte effects on multiple plant vital rates across a broad precipitation gradient under field conditions. This integrative approach provides a more comprehensive assessment of how climate may shape the demographic consequences of plant–microbial symbioses.

Comment 2: *"The introduction establishes potential costs of symbiosis increasing with stress, but the authors' hypotheses ignore this potential. I think it would be helpful to explain why the authors focused on only hypotheses that were majority beneficial before the hypotheses were established."*

Response: We thank Reviewer 1 for pointing out this issue. We agree that our original presentation did not sufficiently explain why our hypotheses were framed primarily around conditions under which endophyte symbiosis provides demographic benefits, despite acknowledging that the costs of maintaining symbionts may increase with environmental stress. Our intention was not to assume that increasing stress necessarily strengthens mutualism, but rather to test whether the well-documented benefits of *Epichloë* symbiosis are maintained, enhanced, or eroded across environmental conditions. We have therefore clarified this rationale in the Introduction and revised the climate-dependent hypothesis to explicitly recognize that endophyte benefits may weaken or become negative when environmental stress is sufficiently severe. We now clarify that the possibility of reduced or reversed benefits under compounded

stress is incorporated into our herbivory–climate-dependent hypothesis (line 110).

Comment 3: *"In the methods section, it would be helpful to include some of the past research on these hosts plant species. Have past studies found that these species largely benefit from the endophyte presence? A brief review of the past results would help the reader understand the expectations."*

Response: We thank Reviewer 1 for this helpful suggestion. We have added a brief review of past research on these host–endophyte associations to the Study system section of the Methods (line 129), noting that prior work generally reports beneficial effects of endophyte presence on host performance, though the magnitude and direction vary among host species, vital rates, and environmental conditions. This context supports our expectation of predominantly beneficial effects and motivates our focus on how these benefits shift with aridity. The added text reads:

> "Previous studies of these associations have generally found beneficial effects on host performance, although effects can vary among host species, vital rates, and environmental conditions (Davitt et al., 2011; Gundel et al., 2020; Rudgers and Swafford, 2009; Yule et al., 2013)."

Comment 4: *"Given the results that endophytes benefit the host only under more mesic conditions, I am left wondering, what are the implications of these results for the symbiosis? Does the proportion of symbiotic hosts decline at the drier ends of the spectrum? I would recommend that the authors touch on some of these biological implications."*

Response: We thank Reviewer 1 for highlighting this important biological implication. We agree that the stronger demographic benefits of endophytes under mesic conditions raise the possibility that the balance of costs and benefits could influence the persistence and distribution of the symbiosis across environmental gradients. We have expanded the Discussion to address this implication. Specifically, we now discuss the possibility that if endophyte-associated costs reduce host or symbiont fitness under dry conditions, endophyte prevalence could decline toward the dry end of environmental gradients, consistent with evidence that endophyte incidence varies substantially across broad environmental gradients (Afkhami and Rudgers, 2009; Semmartin et al., 2015). We emphasize, however, that this is a prediction arising from our demographic results rather than a pattern we directly tested, and that direct studies of endophyte prevalence and transmission across environmental

gradients are needed to determine whether the demographic pattern we observed is associated with changes in symbiosis frequency (line 391).

Comment 5: *"The conclusion states that the endophyte cannot be expelled from the host and this may have implications for the stress gradient hypothesis; however, there is evidence of imperfect transmission of endophytes between tillers. How may this affect the results of the study? Did the authors test the final proportions of symbiont hosts?"*

Response: We thank Reviewer 1 for pointing out this important nuance. We agree that our original statement that vertically transmitted endophytes "cannot be expelled" was too strong. We have revised the text to clarify that, although *Epichloë* endophytes are tightly integrated with their hosts and are not readily removed from established infected tissues, imperfect transmission can result in symbiont loss among tillers and across reproductive transitions. We now explicitly discuss how such transmission dynamics could influence the prevalence and persistence of the symbiosis when the costs of infection increase under stressful conditions (line 389).

Regarding the reviewer's question, we did not re-assess endophyte status at the end of the experiment and therefore cannot determine whether endophyte prevalence changed during the experiment or whether all tillers remained infected. Our analyses therefore assume that the initial symbiont treatment was maintained throughout the experiment. We now state this limitation explicitly and discuss how changes in endophyte prevalence or within-host distribution could affect the interpretation of the demographic responses (line 389).

Comment 6: *"179–181: How did the destruction of these exclusions affect the results? Were these effects included in the models? Given the high variability of the exclusion efficacy (Figure S3), it would be helpful to provide more detail on the implications of the replanting and any effects on the results."*

Response: We thank Reviewer 1 for raising this important point. Because demographic responses were calculated between consecutive censuses, observations from replanting the KER and SON sites were treated as new t–$t+1$ demographic intervals rather than excluded outright, so the disturbance itself does not introduce missing or miscoded transitions (line 194). To directly test whether this disturbance and replanting – together with the variability in exclusion efficacy at these sites shown in Figure S3 – influenced our conclusions, we repeated the complete analy-

ses excluding KER and SON. Posterior estimates of endophyte effects on survival, growth, inflorescence production, and spikelet production were nearly identical between the full dataset and the dataset excluding these two sites across all species, vital rates, and herbivory treatments (Fig. S6, Table S6). Across the 66 species × vital rate × herbivory × precipitation-level combinations evaluated, the estimated direction of the endophyte effect was unchanged in 65/66 cases, and the 11 effects with strong posterior support (P(Δ>0) >0.95 or <0.05) in the full-data analysis retained the same direction and strong support in 10/11 cases when KER and SON were excluded. This indicates that neither the destruction/replanting event nor the associated exclusion-efficacy variability at these sites is driving our main results. We have added this sensitivity analysis, including the full results (Table S6) and figure (Fig. S6) shown below, to the Supplementary Materials, and describe the analysis and its outcome in the Methods (line 194) and Results (line 339) of the revised manuscript.

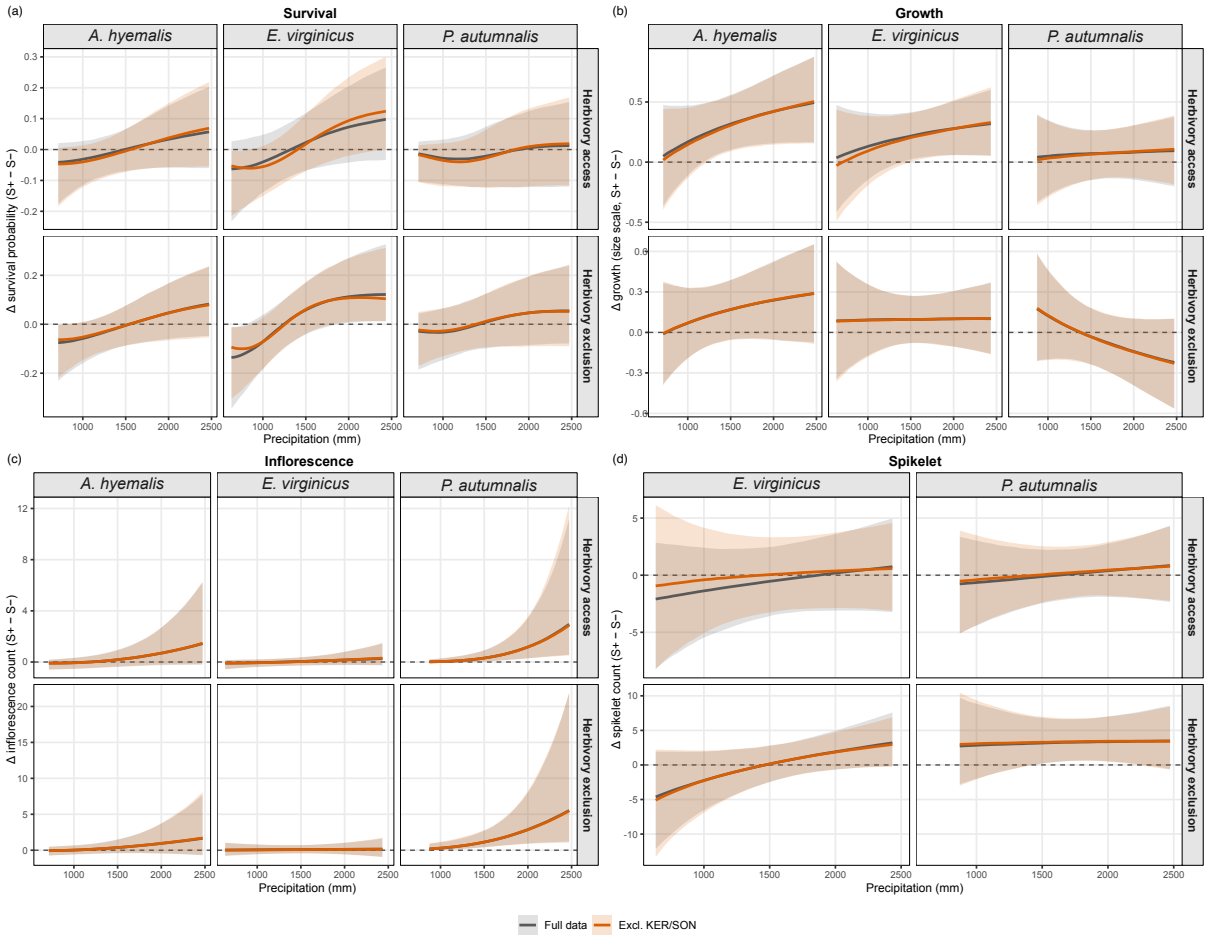


Figure S7: Posterior median (line) and 90% credible interval (shading) of endophyte effects ($\Delta = S^{+} - S^{-}$) for the full dataset (gray) and for the dataset excluding KER and SON (orange), by vital rate, species, and herbivory treatment.

Comment 7: *"190–193: How were the spikelets for A. hyemalis estimated? Table S9 has model results for A. hyemalis but not P. autumnalis."*

Response: Spikelet counts were not recorded for *A. hyemalis* because its inflorescences contain numerous very small spikelets that are difficult to count reliably in the field. We have clarified in the revised Methods (line 206) that spikelets were instead counted directly on up to three inflorescences from reproducing individuals of *P. autumnalis* and *E. virginicus*, while reproduction data in *A. hyemalis* were limited to inflorescence counts. We also corrected the supplementary table accordingly; the relevant table is now Table S10 and reports spikelet model results for *P. autumnalis* and *E. virginicus* only.

Comment 8: *"220: It may be helpful for clarity to replace "endo" with "symb" in the model since the test mostly describes the treatment as S^+ vs. S^-."*

Response: We thank Reviewer 1 for pointing out this inconsistency. We have replaced "endo" with "symb" in the model to consistently represent the S^+ versus S^- treatment throughout the analysis.

Comment 9: *"276: There appears to be a typo here since A. hyemalis appears to experience more herbivore damage by percentage when S^+."*

Response: We thank Reviewer 1 for catching this error. The raw percentages for *A. hyemalis* were reported in an inconsistent order, which made it appear that S^+ plants experienced more herbivore damage. We now report model-based predictions from our binomial model of tiller damage (Fig. S6) for all three species, consistently as S^+ vs. S^-. These show that endophyte symbiosis reduced damage in *P. autumnalis* and, more weakly, in *A. hyemalis*, but increased damage in *E. virginicus* (line 289).

Comment 10: *"310: Cite the figure or table with the spikelet results."*

Response: We thank Reviewer 1 for catching this omission. We have added a citation to Table S10, which reports the spikelet production model results for *P. autumnalis* and *E. virginicus*, at this point in the text (line 324).

Comment 11: *"Figure 1. It is hard to read the y-axis S+ versus S-. Would it be possible to not make them superscript or increase the size?"*

Response: We thank Reviewer 1 for this helpful suggestion. We have revised Figure

1 to improve the readability of the S^+ and S^- labels by increasing their size and presenting them as regular text rather than superscript.

Response to Reviewer 2

Comment 12: *"This manuscript presents results of a field experiment that tests predictions of the classic stress gradient hypothesis (SGH) in plant-endophyte systems. Strengths of this study include testing of clear and biologically interesting hypotheses, rigorous field experiments and statistical analyses, and inclusion of a small complementary greenhouse study to test one particular mechanism that might have confounded their field study findings. Notably, results challenge the predictions of the classics SGH, and the authors propose several explanations for this contradictory finding in the discussion. My main criticism is that the framework for this study seems somewhat over-simplified. There has been substantial work since the original SGH that adds nuance to this concept, especially that SGH predictions may change under extreme or compounding stressors such that positive interactions should increase with decreasing environmental severity (e.g., see Michalet et al. 2006, Kawai and Tokeshi 2007, le Roux and McGeoch 2010). These ideas are not mentioned in the introduction at all and only mentioned indirectly in the discussion. Otherwise, this is a strong, well-written study that should be of broad interest to ecologists interested in positive interactions."*

Response: We thank Reviewer 2 for highlighting this important development of the stress-gradient hypothesis. We agree that our original framing was too narrow in emphasizing the classic prediction of a monotonic increase in positive interactions with increasing stress. We have revised the Introduction to acknowledge that subsequent work has identified important departures from this pattern, including plateaus or declines in mutualism or facilitation under extreme stress and changes in interaction outcomes when multiple stressors occur simultaneously. Specifically, we now note that the intensity, type, and combination of stressors can determine whether increasing stress strengthens or weakens positive interactions, and we cite David et al. (2018); Kawai and Tokeshi (2007); Le Roux and McGeoch (2010); Michalet (2007) in this context (line 47). We have also revised our framing of the hypotheses to recognize that endophyte benefits may weaken or become negative when environmental stress becomes sufficiently severe that the costs of maintaining the symbiosis outweigh its benefits. Thus, although our experimental predictions retain the classic SGH as an important benchmark, we now explicitly consider the possibility of

declining or reversed benefits under greater environmental stress.

Finally, we have revised the Discussion to explicitly interpret our results in light of these extensions of the SGH. We now distinguish our departure from the classic monotonic prediction from the broader SGH literature, noting that our finding of declining endophyte benefits with increasing aridity is consistent with the possibility that positive interactions can weaken or reverse under sufficiently severe stress. We also discuss why these extensions may be particularly relevant to symbioses in which hosts have limited flexibility to regulate symbiont load under stress (line 359). Together, these changes provide a more nuanced conceptual framework for interpreting our finding that endophyte benefits were strongest under mesic rather than arid conditions, without implying that our study directly tests all of the alternative functional forms proposed in the broader SGH literature.

Comment 13: *"One very minor thing to note is that some supplementary figures are not correctly referenced in the text (e.g., line 240 should be Fig. S6 instead of S5 and line 277 should be Fig. S3 instead of S4)."*

Response: We thank Reviewer 2 for carefully checking the supplementary figure references. We have rechecked the manuscript and corrected the supplementary figure citations accordingly (line 279).

Response to Reviewer 3

Comment 14: *"This paper reports a study of symbiotic Epichloe effects on four vital rates of three grass species over a precipitation gradient in the field, with and without herbivore exclusion. The work is supplemented with greenhouse experiments to test (and exclude) possible effects of soil origin. I will qualify my report with the caveat that I'm not well versed in the statistical methods. I feel that this is a very well planned and executed study, and provides important new, hypothesis driven insight into the grass-Epichloe systems specifically, and the principles of symbiosis generally. I have suggestions to improve the manuscript by filling it out with some additional information (especially in Methods), plus some corrections and clarifications."*

Response: We thank Reviewer 3 for the positive assessment of our study and for the detailed and constructive suggestions. We have addressed each of the points raised below and made corresponding revisions to the manuscript.

Comment 15: *"Line 124 and throughout: "E. typhina subsp. poae" should be "E. poae", since it is now recognized as a species distinct from E. typhina. (Also, for taxonomic names, don't italicize words or abbreviations such as sp. and subsp.)"*

Response: We thank Reviewer 3 for noting this taxonomic update. We have revised the manuscript accordingly, replacing *E. typhina* subsp. *poae* with *E. poae* throughout and correcting the formatting of taxonomic abbreviations (e.g., "sp." and "subsp.") as suggested (line 127).

Comment 16: *"Line 124: Do not italicize PauTG-1."*

Response: Corrected as suggested (line 127).

Comment 17: *"Line 127: Disagreement in number: "stroma" is singular. The plural is "stromata"."*

Response: We thank the reviewer for noting this grammatical error. We have corrected "stroma" to "stromata" where the plural form is intended (line 135).

Comment 18: *"Line 136 and throughout: I think it looks better to use a proper minus sign (or n-dash) instead of a hyphen for designations such as S–. (Also, does this journal use n-dash for ranges?)"*

Response: We thank the reviewer for pointing this out. We have replaced the hyphen with a proper minus sign throughout the manuscript for designations such as S^-. We have also followed the journal's typographical convention of using an en dash for numerical ranges (lines 144, 147, 159, 164, 190, 228).

Comment 19: *"Line 137: Should "five days" be "5 d"?"*

Response: Thank you for the suggestion. We have retained "five days" rather than "5 d," as the journal guidelines do not require the abbreviation "d" for days, and we follow the convention of spelling out numbers below 10 in the text.

Comment 20: *"Line 138: Never spell out "minutes". The standard unit is "min"."*

Response: Thank you for the suggestion. We have changed "minutes" with "min" as suggested.

Comment 21: *"Line 142: Give the actual concentration of sodium hypochlorite. Also,*

the surface sterilization procedure was undoubtedly more involved than stated, so either a full description or reference to it in the literature is needed here."

Response: We thank Reviewer 3 for pointing this out. We have revised the Methods to specify that seeds were surface-sterilized using a 10% bleach solution (Sahu et al., 2022) (line 150).

Comment 22: *"Line 143: More detail needed here also: what temperature? What light regime?"*

Response: Our greenhouses maintain a range of 65-85F under ambient light and humidity. We have added these details to the manuscript (line 150).

Comment 23: *"Lines 147–150 (and Table S1): I have a hard time interpreting this and relating it to the experiment as described in the section following. Several questions: What is the operational definition of genotype in this study? How was that diversity established and maintained? What is the relevance of the numbers of genotypes reported here and in Table S1? It seems that, from the information in Lines 174-181 that only a fraction of that genotype diversity was actually utilized in the experiments."*

Response: We thank the reviewer for pointing out this lack of clarity. We have revised the Methods to define "genotype" operationally as an individual derived from a single seed and its vegetatively propagated descendants, such that clones derived from the same genotype represent identical genetic backgrounds for both host and symbiont (line 154). We now clarify that these source genotypes were maintained through vegetative propagation and refer readers to Table S1 for the number of source genotypes and clonal replicates represented in each source population. The relevance of these numbers is so that readers understand that the total replication of the experiment involves genetically identical clones, but to varying degrees for different species and populations. To make this point clearer, we have clarified how these vegetatively propagated plants relate to the plants included in the experiment (line 154).

Comment 24: *"Lines 176–177 "Each plot was planted with 15 individuals, a mix of S^+ and S^- plants": How many of each (S^+ vs S^-)?"*

Response: Thank you for pointing this out. We have clarified the numbers of S^+ and S^- plants corresponding to each starting endophyte frequency (line 179).

Comment 25: *"Line 189: Should "on up to three" be "on up of three"?"*

Response: Thank you for the suggestion. We have rephrased this as "on a maximum of three" to clarify that spikelets were counted on no more than three inflorescences per reproducing individual (line 206).

Comment 26: *"Lines 192-193: I'm not sure of the reason for this sentence. I would think it sufficient to state that A. hyemalis spikelets were too numerous and small to count (maybe a rephrasing of the previous sentence)."*

Response: We thank the reviewer for the suggestion. We have removed this sentence to reduce redundancy and clarified in the preceding sentence that *A. hyemalis* spikelets were too numerous and small to count (line 206).

Comment 27: *"Line 193: Change "while" to "whereas". "While" implies simultaneous occurrence. I suggest you check usage of "while" throughout the paper. Often a better word in context is although, but, whereas, etc."*

Response: Thank you for the suggestion. We have changed "while" to "whereas" here to make the contrast between species more explicit (line 198).

Comment 28: *"Statistical modeling section: I think it might help the reader if the name the vital rates and explicitly connect each to the parameters. In Fig S5 the vital rates are named Survival, Growth, Inflorescence and Spikelet. But line 204 lists "survival, growth, and fertility." Some clarification is needed here."*

Response: Thank you for pointing this out. We have clarified the terminology by specifying that fertility was quantified as inflorescence and spikelet production (line 219).

Comment 29: *"Line 210: Actually, I don't know what the distinction is "between flowering occurrence and inflorescence production". Can this be explained?"*

Response: We agree that this distinction was not sufficiently clear. We have clarified that the hurdle model distinguishes between whether a plant produced any inflorescences and, conditional on producing any inflorescences, how many it produced (line 222).

Comment 30: *"Lines 275-276, "S^+ hosts experienced less herbivore damage than S^-*

hosts in A. hyemalis (19% vs. 15%) and P. autumnalis (13% vs. 18%)": The orders of the numbers given aren't consistent, and would cause me to think that there is more damage on S+ for A. hyemalis, but less on S+ for P. autumnalis."

Response: We thank Reviewer 3 for pointing out this inconsistency. We agree that the order of the values was confusing, and we have corrected it. All values are now model-based predictions from our tiller damage model (Fig. S6), with each pair explicitly labelled as S^+ vs. S^- (line 289).

Comment 31: *"Lines 344-345, "These results contradict predictions...": Is the contradiction with respect to the abiotic stresses only, or is the herbivore effect also in contradiction of the hypothesis? The context (previous sentence) seems to imply the latter, but I doubt that's what's meant."*

Response: We thank the reviewer for identifying this ambiguity. We agree that our original wording could be interpreted as suggesting that the herbivory results also contradicted the SGH. This was not our intent. Our departure from the classic SGH prediction concerns the abiotic stress gradient, where endophyte benefits declined rather than increased with increasing aridity. In contrast, herbivory effects were inconsistent across species and fitness components, although the strongest effects were in the predicted direction. We have revised the text to make this distinction explicit and to clarify that our results provide mixed evidence for the predicted effects of herbivory (line 359).

Comment 32: *"Lines 385-388: It should also be noted that the endophytes typically associated with these grasses may differ strongly in which and how much anti-herbivore alkaloids they have. For example, high levels of the potent anti-insect loline alkaloids (plus low levels of peramine) have been documented in Poa autumnalis, whereas Elymus virginicus has been found to have chanoclavine (activity unknown) and peramine (a feeding deterrent, not as potent as lolines), and different chemotypes are reported for different E. amarillans isolates from A. hyemalis. See the cited alkaloid literature, but note that especially for sexual Epichloe species such as in E. virginicus and A. hyemalis, there is likely to be considerable variation. In Poa autumnalis (Siegel et al. 1990), which has an asexual endophyte, one might expect a consistent alkaloid profile compared to what has been published, but there is no guarantee."*

Response: We thank the reviewer for this important point. We agree that differences among the endophytes in the identity and abundance of anti-herbivore alkaloids could contribute to the species-specific herbivory responses we observed. We

have revised the Discussion to acknowledge this possibility and added the relevant literature on alkaloid profiles in the endophytes associated with *P. autumnalis*, *E. virginicus*, and *A. hyemalis*. Because we did not characterize endophyte strains or quantify alkaloid profiles in this study, we now present variation in endophyte chemistry as a potential explanation rather than attributing the observed responses to specific alkaloids. We also retain host condition as a potential explanation for the increased herbivory observed in endophyte-symbiotic *E. virginicus*, but have revised the text to make clear that these mechanisms are not mutually exclusive (line 421).

Comment 33: *"Line 403: I didn't see where this was tested. It seems that the underlying assumption was that the S^+ plants retained the endophyte throughout the experiment. Were there any data taken on endophyte presence later, such as at the end of the study?"*

Response: We thank the reviewer for pointing out this important limitation. We agree that our study did not directly test whether hosts retained their endophytes throughout the experiment, as we did not re-assess endophyte presence at the end of the study. We have therefore revised the Discussion to frame host control over symbiont retention as a potential mechanism rather than a conclusion supported by our data, and we now explicitly acknowledge that we cannot determine whether variation in symbiont retention contributed to the demographic patterns we observed (line 391).

Comment 34: *"Lines 417-420: Can the authors comment—even speculate— on why other studies have found host-range expansion effects of Epichloe? Does that have to do with specifically which factors limit the hosts and establish those "range margins"?"*

Response: We thank the reviewer for this insightful suggestion. Although we did not directly test why previous studies have found host-range expansion effects of *Epichloë*, we now speculate that these contrasting outcomes may depend on the environmental factors establishing host range margins. We have added this possibility to the Discussion, noting that symbiosis may promote host persistence or range expansion when endophyte benefits offset the factor limiting host performance, but provide little benefit or even constrain performance when the limiting factor is severe resource stress that the symbiosis cannot alleviate (line 434).

Comment 35: *"Figures: In general I think that saturation contrast can be used to better*

distinguish treatments in line graphs and bar graphs (Fig. 1, Fig. 2 panels b, d and f, Fig. 5, and Fig. 6 panel e). In general, I suggest checking readability for common types of color blindness."

Response: We thank the reviewer for this helpful suggestion. We revised the color palettes across all figures to increase saturation and contrast between treatments and improve visual distinction. We also adopted a consistent cool-toned, colorblind-accessible palette throughout the figures and checked the revised figures for readability under common forms of color-vision deficiency.

Comment 36: *"Fig. S2 and S3: I think the bar colors and orders, viz herbivore access vs exclusion, are switched between these two figures. That's confusing. (Also try to match color usage in Figs. 1, 5, and S10)."*

Response: We have corrected the color/order inconsistency between Figs. S2 and S3 (now S4 and S5) and matched color usage across Figs. 1, 5, and S10.

Comment 37: *"Figs. S5 and S6: What are the axes?"*

Response: We thank the reviewer for noting the need for clearer axis labels. We have revised the posterior predictive check figure to explicitly identify the response variable on the x-axis for each vital rate (survival status, growth, inflorescence count, and spikelet count) and to label the y-axis as probability density, which represents the density of the observed and posterior predictive distributions. We also added panel labels (A–D) to facilitate reference to individual vital rates.

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Supporting Information

Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

Jacob K. Moutouama, Joshua C. Fowler, Akiem M. Gough, Julia Martin, Christopher C. Oxley, Karl Schrader, Karis Williams, and Tom E.X. Miller

S.1 Supporting Information

S.1.1 Supporting Methods

S.1.1.1 Endophyte status verification

To assess the efficacy of heat treatment and to confirm endophyte status before transfer to the field experiment, we verified symbiont status (S^+ or S^-) of plants from control and heat lineages using three methods. First, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify *Epichloë* hyphae (**?**). Second, we used microscopy to assess the presence or absence of *Epichloë* in seeds that were soaked in NaOH, then squashed and stained similarly to the leaf tissue, to determine the endophyte status of the maternal plant (Fowler et al., 2025). Third, we used an immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co., Watkinsville, Georgia, USA) that targets proteins of *Epichloë* spp. to visually indicate presence or absence (**?**). In total, we tested 1,481 plants for endophyte status. In cases where two or more methods were used on the same plants ($N = 253$), there was 99.2% agreement between methods. For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assumed that its status applied equally to all clones.

All S^- plants used in the experiment came from heat-treated lineages, while S^+ plants came from both control and heat lineages depending on the efficacy of the removal treatment. While heat treatment was effective at reducing endophyte prevalence relative to controls, our confidence that individual S^- plants were experimentally disinfected, as opposed to naturally symbiont-free, varied due to among-population variation in endophyte prevalence (mean: 68% S^+) and the effectiveness of the heat treatment (Table **??**). For example, for the SJC population of *A. hyemalis*, control and heat lineages yielded plants that were 100% and 5% S^+, respectively – giving us high confidence that S^- plants are derived from successful symbiont removal. For the SFAEF population of *P. autumnalis*, on the other hand, control and heat lineages yielded plants that were 35% and 26% S^+, respectively – meaning that these S^- plants were quite likely from naturally S^- lineages.

S.1.2 Greenhouse experiment

S.1.2.1 Experimental design

To examine whether edaphic factors mediate plant performance across the precipitation gradient, we conducted a greenhouse experiment manipulating soil origin and endophyte status. Soils were collected during summer 2024 from the seven field sites used in the range-limits experiment. These sites span the precipitation gradient from the wettest site to the driest site. After collection, soils were stored in a cold room at 3.3°C until the start of the experiment. Soils were not sterilized and therefore soil origin treatment reflects both the physical properties of the soil as well as the microbial communities that may vary across soil types.

Seeds of *Agrostis hyemalis* were obtained from the seed stock used to generate plants in the range-limits field experiment. Seeds originated from two source populations (LPEF and SJC) and were grouped according to endophyte status as endophyte-symbiotic (S^+) or endophyte-free (S^-). Seeds were germinated on agar plates (18.2 g agar per 1000 mL deionized water) in a growth chamber and subsequently transplanted as seedlings into pots containing soils from each field site. Plants were randomly assigned to greenhouse trays and maintained under controlled greenhouse conditions with automated irrigation. The experimental design crossed seed source population, endophyte status, and soil origin. Realised sample sizes varied across treatment combinations due to mortality and poor germination and are reported in Table **??**.

S.1.2.2 Data collection

Plants were monitored regularly for survival and reproductive development. Reproductive output was quantified as inflorescence number and spikelet production. During the flowering season, plants were checked at least twice per week for the presence of seed-bearing inflorescences. Mature inflorescences were collected as they developed to avoid repeated counts. For each flowering individual, spikelets were counted on at least three inflorescences when available. For individuals producing many inflorescences, mean spikelet number per inflorescence was used to estimate total spikelet production. After eight months of growth, plants were harvested to quantify aboveground biomass. Plant material was dried at 60°C in a drying oven and weighed using an analytical balance. Because inflorescences were collected prior to harvest, they were dried separately and added to the total aboveground biomass.

S.1.2.3 Statistical analysis

For each fitness component (aboveground biomass, inflorescence count, and spikelets per inflorescence), we fitted hierarchical Bayesian models mirroring the field analysis. Fixed

effects included symbiont status (S^+/S^-), the 30-year mean annual precipitation of each soil's origin site (standardised; hereafter soil-origin MAP), and their interaction, with planting date (standardised) as a covariate. Source population was included as a random effect with a non-centred parameterisation. Biomass was log-transformed and modelled with a Student-t likelihood ($\nu = 3$) to provide robustness to outliers, which are common in small greenhouse experiments; count responses (inflorescence number and spikelets per inflorescence) used negative binomial likelihoods to accommodate overdispersion. For all models, fixed-effect coefficients received Normal$(0,1)$ priors; the overdispersion parameter for count models was placed on the log scale with a Normal$(2,1)$ prior (centred on $\phi \approx 7$); and the random-effect standard deviation received an Exponential(2) prior.

Models were fitted in Stan (Stan Development Team, 2024) with four chains of 2,000 iterations (1,000 warmup; adapt_delta $= 0.99$). Convergence was assessed via $\hat{R} < 1.01$ for all parameters and inspection of trace plots. We report posterior medians with 95% credible intervals for three focal parameters: the symbiont main effect (S^+ vs. S^- at mean soil-origin MAP), the soil-origin MAP main effect, and the symbiont $\times$ soil-origin MAP interaction. To visualise how the endophyte effect varies across the soil-origin precipitation gradient, we computed the posterior contrast ($S^+ - S^-$) at each point along the observed precipitation range, marginalising over planting date (set to the mean) and source population random effects. These contrasts are summarised as posterior medians with 95% credible intervals and displayed alongside the predicted responses for each symbiont status. Posterior predictive checks indicated adequate fit for all models (Fig. **??**).

Supporting Tables

Table S1: Geographic coordinates and descriptions of source populations for study species. For each source population, the table shows the total number of plants included in the experiment, the number of unique genotypes, and the average clonal replication per genotype.

Species	Population	Latitude	Longitude	Notes	Total plants	Unique genotypes	Mean clones per genotype
A. hyemalis	SJC	30.7720	−95.2006	Private property, San Jacinto County, TX	273	201	1.36
A. hyemalis	COHR	29.6728	−96.5675	Roadside collection, Columbus, TX	289	35	8.26
A. hyemalis	LPEF	30.0854	−97.1725	Stengl Lost Pines Field Station, Bastrop, TX	278	41	6.78
P. autumnalis	LA7	30.8497	−93.2768	Roadside population, DeRidder, LA	91	9	10.11
P. autumnalis	LA1	32.5681	−93.9330	Walter P. Jacobs Nature Center, Shreveport, LA	129	7	18.43
P. autumnalis	LA2	31.1830	−92.6170	Kisatchie National Forest, LA	253	15	16.87
P. autumnalis	SFAEF	31.4922	−94.7722	Stephen F. Austin Experimental Forest, TX	247	41	6.02
E. virginicus	HUNT	30.7418	−95.4745	Pineywoods Environmental Research Laboratory, Huntsville, TX	91	61	1.49
E. virginicus	RL5	30.4717	−99.7838	Llano River Field Station, Junction, TX	131	91	1.44
E. virginicus	PALM	29.5916	−97.5848	Palmetto State Park, TX	442	195	2.27
E. virginicus	JLP	29.7851	−90.1149	Jean Lafitte Park, LA	176	102	1.73

Table S2: Summary of symbiont removal treatments by species and population. N indicates the number of plants tested for symbiont status and $\%S^+$ indicates the proportion that tested positive. Treatment refers to the symbiont removal treatment applied to the seed lineage: Control (no treatment) or Heat (5 days in a drying oven at 60°C for *E. virginicus* and *P. autumnalis*; 12 min in a 60°C water bath for *A. hyemalis*). For all populations except SFAEF and PALM, heat-treated seed lineages were one or more generations removed from the treatment; for those two populations, plants from heat-treated seeds were used directly in the experiment.

Species	Population	Treatment	N	%S+
A. hyemalis	COHR	Control	84	0.92
A. hyemalis	COHR	Heat	75	0.04
A. hyemalis	LPEF	Control	7	0.57
A. hyemalis	LPEF	Heat	88	0.00
A. hyemalis	SJC	Control	132	1.00
A. hyemalis	SJC	Heat	191	0.05
E. virginicus	HUNT	Control	71	0.61
E. virginicus	HUNT	Heat	38	0.21
E. virginicus	JLP	Control	27	1.00
E. virginicus	JLP	Heat	113	0.02
E. virginicus	PALM	Control	176	0.61
E. virginicus	PALM	Heat	238	0.71
E. virginicus	RL5	Control	68	0.43
E. virginicus	RL5	Heat	74	0.73
P. autumnalis	LA1	Control	5	0.20
P. autumnalis	LA1	Heat	4	0.00
P. autumnalis	LA2	Control	10	0.80
P. autumnalis	LA2	Heat	3	0.33
P. autumnalis	LA7	Control	5	1.00
P. autumnalis	LA7	Heat	11	0.00
P. autumnalis	SFAEF	Control	23	0.35
P. autumnalis	SFAEF	Heat	38	0.26

Table S3: Common garden sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
Private property, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S4: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOO-CV). ΔELPD is relative to the best model.

Vital rate	Model type	ΔELPD	SE
Survival	Linear	0.000	0.000
Survival	Quadratic	-0.282	0.663
Growth	Linear	0.000	0.000
Growth	Quadratic	-0.078	0.430
Inflorescence	Linear	0.000	0.000
Inflorescence	Quadratic	-0.689	0.516
Spikelet	Linear	0.000	0.000
Spikelet	Quadratic	-0.332	0.579

Table S5: Sample sizes for each combination of seed source population, soil-origin site, and endophyte treatment in the greenhouse experiment. The experiment crossed two seed source populations (LPEF and SJC), seven soil-origin sites along the precipitation gradient, and two symbiont treatments (S^- and S^+). Most treatment combinations included seven replicates, except LPEF $\times$ BFL $\times$ S^- where $n = 6$ due to early mortality.

Response	Site	S-	S+
Biomass			
	LAF	14	14
	SON	14	14
	KER	14	14
	BFL	13	14
	BAS	14	14
	COL	14	14
	HUN	14	14
Inflorescence			
	LAF	14	14
	SON	14	14
	KER	14	14
	BFL	14	14
	BAS	14	14
	COL	14	14
	HUN	14	14
Spikelets			
	LAF	7	10
	SON	3	1
	KER	1	0
	BFL	3	8
	BAS	7	6
	COL	2	2
	HUN	5	12

Table S6: Sensitivity of estimated endophyte effects to exclusion of sites KER and SON, which were destroyed by deer and subsequently replanted in 2024. For each combination of vital rate (Trait), species, herbivory treatment (Access = herbivory-access plots, Exclusion = herbivore-exclusion plots), and precipitation (Precip., in mm; three values spanning each species' observed precipitation range), the table reports the posterior median and 90% credible interval [Med. Δ (90% CI)] and the posterior probability of a positive endophyte effect [P(Δ > 0)] for the full dataset and for the dataset excluding KER and SON. The endophyte effect Δ is defined as Pr(S+) − Pr(S-) for survival, or the analogous difference on the response scale (S+ − S-) for growth, inflorescence production, and spikelet production. Spikelet production was not modeled for *A. hyemalis* because spikelets in this species were too numerous and small to count reliably (see Methods). Posterior estimates were nearly identical between the full dataset and the dataset excluding KER and SON across all species, vital rates, and herbivory treatments, indicating that the disturbance and replanting at these two sites did not meaningfully influence our conclusions.

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Growth	*A. hyemalis*	Access	714	0.05 [-0.37, 0.47]	0.58	0.02 [-0.40, 0.44]	0.53
Growth	*A. hyemalis*	Access	1300	0.27 [0.05, 0.51]	0.98	0.25 [0.03, 0.50]	0.97
Growth	*A. hyemalis*	Access	2472	0.49 [0.16, 0.88]	0.99	0.50 [0.16, 0.88]	0.99
Growth	*A. hyemalis*	Exclusion	714	-0.01 [-0.38, 0.37]	0.48	-0.01 [-0.39, 0.38]	0.49
Growth	*A. hyemalis*	Exclusion	1300	0.14 [-0.08, 0.35]	0.85	0.13 [-0.09, 0.35]	0.84
Growth	*A. hyemalis*	Exclusion	2472	0.29 [-0.07, 0.65]	0.90	0.29 [-0.09, 0.65]	0.90
Growth	*E. virginicus*	Access	633	0.04 [-0.41, 0.47]	0.56	-0.03 [-0.49, 0.44]	0.46
Growth	*E. virginicus*	Access	1213	0.17 [-0.05, 0.40]	0.90	0.15 [-0.09, 0.38]	0.85
Growth	*E. virginicus*	Access	2433	0.32 [0.05, 0.60]	0.97	0.33 [0.06, 0.62]	0.98
Growth	*E. virginicus*	Exclusion	633	0.09 [-0.34, 0.53]	0.63	0.08 [-0.37, 0.52]	0.62
Growth	*E. virginicus*	Exclusion	1213	0.10 [-0.11, 0.30]	0.78	0.09 [-0.12, 0.30]	0.77
Growth	*E. virginicus*	Exclusion	2433	0.10 [-0.16, 0.37]	0.74	0.10 [-0.16, 0.37]	0.73
Growth	*P. autumnalis*	Access	873	0.04 [-0.33, 0.40]	0.57	0.02 [-0.36, 0.39]	0.53
Growth	*P. autumnalis*	Access	1443	0.07 [-0.16, 0.27]	0.70	0.06 [-0.16, 0.26]	0.69
Growth	*P. autumnalis*	Access	2472	0.10 [-0.20, 0.38]	0.70	0.11 [-0.19, 0.39]	0.74
Growth	*P. autumnalis*	Exclusion	873	0.17 [-0.21, 0.58]	0.78	0.18 [-0.21, 0.59]	0.77

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Table S6 – continued from previous page

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Growth	*P. autumnalis*	Exclusion	1443	-0.02 [-0.22, 0.20]	0.45	-0.02 [-0.23, 0.20]	0.45
Growth	*P. autumnalis*	Exclusion	2472	-0.22 [-0.56, 0.10]	0.13	-0.23 [-0.56, 0.10]	0.13
Inflorescence	*A. hyemalis*	Access	714	-0.09 [-0.59, 0.18]	0.25	-0.09 [-0.60, 0.17]	0.23
Inflorescence	*A. hyemalis*	Access	1300	0.06 [-0.45, 0.62]	0.59	0.05 [-0.42, 0.58]	0.58
Inflorescence	*A. hyemalis*	Access	2472	1.44 [-0.21, 6.22]	0.92	1.47 [-0.14, 6.27]	0.93
Inflorescence	*A. hyemalis*	Exclusion	714	-0.07 [-0.74, 0.49]	0.38	-0.06 [-0.72, 0.49]	0.40
Inflorescence	*A. hyemalis*	Exclusion	1300	0.20 [-0.47, 1.05]	0.71	0.20 [-0.46, 1.01]	0.71
Inflorescence	*A. hyemalis*	Exclusion	2472	1.66 [-0.69, 7.73]	0.88	1.70 [-0.68, 8.07]	0.88
Inflorescence	*E. virginicus*	Access	633	-0.09 [-0.58, 0.17]	0.25	-0.08 [-0.55, 0.14]	0.22
Inflorescence	*E. virginicus*	Access	1213	-0.01 [-0.33, 0.26]	0.46	-0.03 [-0.32, 0.24]	0.43
Inflorescence	*E. virginicus*	Access	2433	0.27 [-0.25, 1.44]	0.82	0.31 [-0.25, 1.50]	0.83
Inflorescence	*E. virginicus*	Exclusion	633	0.03 [-0.81, 1.06]	0.53	0.01 [-0.74, 1.02]	0.52
Inflorescence	*E. virginicus*	Exclusion	1213	0.07 [-0.49, 0.72]	0.59	0.06 [-0.48, 0.71]	0.58
Inflorescence	*E. virginicus*	Exclusion	2433	0.15 [-0.91, 1.62]	0.63	0.15 [-0.96, 1.73]	0.63
Inflorescence	*P. autumnalis*	Access	873	0.01 [-0.07, 0.20]	0.66	0.01 [-0.07, 0.21]	0.65
Inflorescence	*P. autumnalis*	Access	1443	0.26 [0.01, 0.78]	0.96	0.25 [0.01, 0.80]	0.96
Inflorescence	*P. autumnalis*	Access	2472	2.95 [0.56, 11.20]	0.99	2.87 [0.52, 12.25]	0.99
Inflorescence	*P. autumnalis*	Exclusion	873	0.18 [0.01, 0.89]	0.97	0.19 [0.01, 0.96]	0.97
Inflorescence	*P. autumnalis*	Exclusion	1443	1.00 [0.36, 2.44]	1.00	1.00 [0.36, 2.52]	1.00
Inflorescence	*P. autumnalis*	Exclusion	2472	5.52 [1.19, 21.88]	1.00	5.48 [1.05, 21.92]	1.00
Spikelet	*E. virginicus*	Access	633	-2.09 [-8.18, 2.83]	0.24	-0.94 [-8.27, 6.13]	0.41

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Table S6 – continued from previous page

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Spikelet	*E. virginicus*	Access	1213	-1.00 [-4.53, 2.31]	0.30	-0.19 [-4.01, 3.60]	0.47
Spikelet	*E. virginicus*	Access	2433	0.74 [-3.23, 4.96]	0.62	0.57 [-3.12, 4.57]	0.60
Spikelet	*E. virginicus*	Exclusion	633	-4.63 [-12.13, 1.94]	0.12	-5.10 [-13.33, 2.20]	0.11
Spikelet	*E. virginicus*	Exclusion	1213	-1.12 [-4.49, 2.23]	0.28	-1.12 [-4.60, 2.25]	0.28
Spikelet	*E. virginicus*	Exclusion	2433	3.20 [-0.17, 7.60]	0.94	2.96 [-0.25, 6.88]	0.93
Spikelet	*P. autumnalis*	Access	873	-0.76 [-5.08, 3.35]	0.35	-0.52 [-5.12, 3.90]	0.42
Spikelet	*P. autumnalis*	Access	1443	-0.18 [-2.60, 2.28]	0.45	-0.01 [-2.58, 2.58]	0.50
Spikelet	*P. autumnalis*	Access	2472	0.84 [-2.35, 4.34]	0.67	0.78 [-2.25, 4.28]	0.67
Spikelet	*P. autumnalis*	Exclusion	873	2.73 [-2.78, 9.74]	0.81	2.99 [-3.03, 10.46]	0.80
Spikelet	*P. autumnalis*	Exclusion	1443	3.13 [0.02, 6.92]	0.95	3.28 [0.05, 7.07]	0.95
Spikelet	*P. autumnalis*	Exclusion	2472	3.46 [-0.60, 8.59]	0.92	3.45 [-0.69, 8.42]	0.92
Survival	*A. hyemalis*	Access	714	-0.04 [-0.18, 0.02]	0.13	-0.05 [-0.18, 0.01]	0.07
Survival	*A. hyemalis*	Access	1300	-0.01 [-0.08, 0.05]	0.35	-0.02 [-0.09, 0.04]	0.26
Survival	*A. hyemalis*	Access	2472	0.06 [-0.06, 0.20]	0.81	0.07 [-0.05, 0.22]	0.84
Survival	*A. hyemalis*	Exclusion	714	-0.07 [-0.23, -0.00]	0.04	-0.06 [-0.22, -0.00]	0.05
Survival	*A. hyemalis*	Exclusion	1300	-0.03 [-0.11, 0.04]	0.25	-0.03 [-0.10, 0.04]	0.26
Survival	*A. hyemalis*	Exclusion	2472	0.08 [-0.05, 0.24]	0.86	0.08 [-0.05, 0.24]	0.84
Survival	*E. virginicus*	Access	633	-0.06 [-0.23, 0.03]	0.11	-0.05 [-0.21, -0.00]	0.03
Survival	*E. virginicus*	Access	1213	-0.01 [-0.10, 0.07]	0.38	-0.03 [-0.11, 0.04]	0.23
Survival	*E. virginicus*	Access	2433	0.10 [-0.03, 0.27]	0.89	0.12 [0.00, 0.30]	0.95
Survival	*E. virginicus*	Exclusion	633	-0.14 [-0.34, -0.01]	0.03	-0.09 [-0.30, -0.01]	0.02

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				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Survival	*E. virginicus*	Exclusion	1213	-0.01 [-0.11, 0.09]	0.44	-0.01 [-0.11, 0.09]	0.41
Survival	*E. virginicus*	Exclusion	2433	0.12 [0.01, 0.33]	0.97	0.10 [0.01, 0.31]	0.98
Survival	*P. autumnalis*	Access	714	-0.01 [-0.10, 0.03]	0.20	-0.02 [-0.10, 0.01]	0.14
Survival	*P. autumnalis*	Access	1300	-0.03 [-0.12, 0.05]	0.26	-0.04 [-0.12, 0.04]	0.19
Survival	*P. autumnalis*	Access	2472	0.01 [-0.12, 0.15]	0.61	0.02 [-0.12, 0.17]	0.65
Survival	*P. autumnalis*	Exclusion	714	-0.03 [-0.19, 0.04]	0.19	-0.02 [-0.17, 0.05]	0.23
Survival	*P. autumnalis*	Exclusion	1300	-0.01 [-0.11, 0.10]	0.41	-0.01 [-0.10, 0.09]	0.44
Survival	*P. autumnalis*	Exclusion	2472	0.06 [-0.08, 0.24]	0.78	0.05 [-0.09, 0.24]	0.77

Table S7: Posterior summaries of Δ (S+ − S−) for survival across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect. Herbivore treatment: Exclusion = herbivores excluded; Access = herbivores had access.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
A. hyemalis	Access	714.00	-0.04	-0.18	0.02	0.13
A. hyemalis	Access	963.00	-0.03	-0.12	0.03	0.16
A. hyemalis	Access	1300.00	-0.01	-0.08	0.05	0.35
A. hyemalis	Access	1832.00	0.02	-0.06	0.12	0.70
A. hyemalis	Access	2472.00	0.06	-0.06	0.20	0.81
A. hyemalis	Exclusion	714.00	-0.07	-0.23	-0.00	0.04
A. hyemalis	Exclusion	963.00	-0.06	-0.17	0.01	0.07
A. hyemalis	Exclusion	1300.00	-0.03	-0.11	0.04	0.25
A. hyemalis	Exclusion	1832.00	0.03	-0.07	0.14	0.70
A. hyemalis	Exclusion	2472.00	0.08	-0.05	0.24	0.86
E. virginicus	Access	525.00	-0.06	-0.27	0.02	0.10
E. virginicus	Access	760.00	-0.06	-0.20	0.03	0.13
E. virginicus	Access	1101.00	-0.03	-0.12	0.06	0.29
E. virginicus	Access	1680.00	0.04	-0.06	0.15	0.77
E. virginicus	Access	2433.00	0.10	-0.03	0.27	0.89
E. virginicus	Exclusion	525.00	-0.13	-0.40	-0.01	0.02
E. virginicus	Exclusion	760.00	-0.12	-0.29	-0.01	0.04
E. virginicus	Exclusion	1101.00	-0.04	-0.14	0.06	0.24
E. virginicus	Exclusion	1680.00	0.09	-0.02	0.21	0.91
E. virginicus	Exclusion	2433.00	0.12	0.01	0.33	0.97
P. autumnalis	Access	714.00	-0.01	-0.10	0.03	0.20
P. autumnalis	Access	963.00	-0.03	-0.11	0.03	0.19
P. autumnalis	Access	1300.00	-0.03	-0.12	0.05	0.26
P. autumnalis	Access	1832.00	-0.00	-0.12	0.11	0.49
P. autumnalis	Access	2472.00	0.01	-0.12	0.15	0.61
P. autumnalis	Exclusion	714.00	-0.03	-0.19	0.04	0.19
P. autumnalis	Exclusion	963.00	-0.03	-0.15	0.06	0.24
P. autumnalis	Exclusion	1300.00	-0.01	-0.11	0.10	0.41
P. autumnalis	Exclusion	1832.00	0.04	-0.09	0.17	0.70
P. autumnalis	Exclusion	2472.00	0.06	-0.08	0.24	0.78

Table S8: Posterior summaries of Δ ($S^+ - S^-$) for growth across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
A. hyemalis	Access	714.00	0.05	-0.37	0.47	0.58
A. hyemalis	Access	963.00	0.15	-0.14	0.47	0.81
A. hyemalis	Access	1300.00	0.27	0.05	0.51	0.98
A. hyemalis	Access	1832.00	0.39	0.15	0.66	1.00
A. hyemalis	Access	2472.00	0.49	0.16	0.88	0.99
A. hyemalis	Exclusion	714.00	-0.01	-0.38	0.37	0.48
A. hyemalis	Exclusion	963.00	0.06	-0.21	0.33	0.65
A. hyemalis	Exclusion	1300.00	0.14	-0.08	0.35	0.85
A. hyemalis	Exclusion	1832.00	0.22	-0.04	0.48	0.91
A. hyemalis	Exclusion	2472.00	0.29	-0.07	0.65	0.90
E. virginicus	Access	525.00	-0.00	-0.53	0.51	0.49
E. virginicus	Access	760.00	0.07	-0.31	0.44	0.64
E. virginicus	Access	1101.00	0.15	-0.10	0.40	0.85
E. virginicus	Access	1680.00	0.24	0.04	0.44	0.98
E. virginicus	Access	2433.00	0.32	0.05	0.60	0.97
E. virginicus	Exclusion	525.00	0.09	-0.42	0.61	0.60
E. virginicus	Exclusion	760.00	0.09	-0.27	0.45	0.65
E. virginicus	Exclusion	1101.00	0.09	-0.14	0.33	0.75
E. virginicus	Exclusion	1680.00	0.10	-0.08	0.28	0.82
E. virginicus	Exclusion	2433.00	0.10	-0.16	0.37	0.74
P. autumnalis	Access	873.00	0.04	-0.33	0.40	0.57
P. autumnalis	Access	1122.00	0.06	-0.23	0.31	0.63
P. autumnalis	Access	1443.00	0.07	-0.16	0.27	0.70
P. autumnalis	Access	1923.00	0.08	-0.14	0.29	0.73
P. autumnalis	Access	2472.00	0.10	-0.20	0.38	0.70
P. autumnalis	Exclusion	873.00	0.17	-0.21	0.58	0.78
P. autumnalis	Exclusion	1122.00	0.08	-0.19	0.37	0.68
P. autumnalis	Exclusion	1443.00	-0.02	-0.22	0.20	0.45
P. autumnalis	Exclusion	1923.00	-0.13	-0.36	0.11	0.19
P. autumnalis	Exclusion	2472.00	-0.22	-0.56	0.10	0.13

Table S9: Posterior summaries of Δ ($S^+ - S^-$) for inflorescence production across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
A. hyemalis	Access	714.00	-0.08	-0.61	0.18	0.26
A. hyemalis	Access	963.00	-0.06	-0.54	0.30	0.37
A. hyemalis	Access	1300.00	0.06	-0.44	0.57	0.59
A. hyemalis	Access	1832.00	0.46	-0.31	1.85	0.85
A. hyemalis	Access	2472.00	1.30	-0.29	6.06	0.91
A. hyemalis	Exclusion	714.00	-0.06	-0.74	0.54	0.40
A. hyemalis	Exclusion	963.00	0.01	-0.63	0.69	0.51
A. hyemalis	Exclusion	1300.00	0.19	-0.50	1.07	0.70
A. hyemalis	Exclusion	1832.00	0.67	-0.49	2.62	0.84
A. hyemalis	Exclusion	2472.00	1.48	-0.95	7.71	0.85
E. virginicus	Access	525.00	-0.09	-0.67	0.16	0.24
E. virginicus	Access	760.00	-0.08	-0.48	0.18	0.29
E. virginicus	Access	1101.00	-0.04	-0.35	0.23	0.41
E. virginicus	Access	1680.00	0.07	-0.24	0.48	0.67
E. virginicus	Access	2433.00	0.25	-0.27	1.36	0.81
E. virginicus	Exclusion	525.00	0.03	-0.87	1.17	0.54
E. virginicus	Exclusion	760.00	0.04	-0.68	0.92	0.56
E. virginicus	Exclusion	1101.00	0.07	-0.52	0.74	0.59
E. virginicus	Exclusion	1680.00	0.09	-0.45	0.75	0.63
E. virginicus	Exclusion	2433.00	0.11	-0.97	1.40	0.60
P. autumnalis	Access	873.00	0.02	-0.07	0.20	0.68
P. autumnalis	Access	1122.00	0.07	-0.06	0.38	0.82
P. autumnalis	Access	1443.00	0.26	0.01	0.83	0.96
P. autumnalis	Access	1923.00	0.98	0.24	2.89	0.99
P. autumnalis	Access	2472.00	2.91	0.48	11.89	0.99
P. autumnalis	Exclusion	873.00	0.19	0.02	1.00	0.97
P. autumnalis	Exclusion	1122.00	0.44	0.10	1.42	1.00
P. autumnalis	Exclusion	1443.00	0.97	0.35	2.44	1.00
P. autumnalis	Exclusion	1923.00	2.37	0.75	6.35	1.00
P. autumnalis	Exclusion	2472.00	4.97	0.91	19.92	0.99

Table S10: Posterior summaries of Δ ($S^+ - S^-$) for spikelet production across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
E. virginicus	Access	525.00	-2.33	-9.38	3.12	0.23
E. virginicus	Access	760.00	-1.84	-7.08	2.69	0.24
E. virginicus	Access	1101.00	-1.19	-4.99	2.31	0.28
E. virginicus	Access	1680.00	-0.27	-3.26	2.81	0.44
E. virginicus	Access	2433.00	0.74	-3.23	4.96	0.62
E. virginicus	Exclusion	525.00	-5.51	-14.68	1.98	0.11
E. virginicus	Exclusion	760.00	-3.70	-9.87	1.93	0.14
E. virginicus	Exclusion	1101.00	-1.69	-5.49	2.03	0.23
E. virginicus	Exclusion	1680.00	0.83	-1.56	3.36	0.71
E. virginicus	Exclusion	2433.00	3.20	-0.17	7.60	0.94
P. autumnalis	Access	873.00	-0.76	-5.08	3.35	0.35
P. autumnalis	Access	1122.00	-0.52	-3.80	2.75	0.38
P. autumnalis	Access	1443.00	-0.18	-2.60	2.28	0.45
P. autumnalis	Access	1923.00	0.31	-1.88	2.57	0.59
P. autumnalis	Access	2472.00	0.84	-2.35	4.34	0.67
P. autumnalis	Exclusion	873.00	2.73	-2.78	9.74	0.81
P. autumnalis	Exclusion	1122.00	2.96	-1.31	8.23	0.88
P. autumnalis	Exclusion	1443.00	3.13	0.02	6.92	0.95
P. autumnalis	Exclusion	1923.00	3.35	0.62	6.84	0.98
P. autumnalis	Exclusion	2472.00	3.46	-0.60	8.59	0.92

Model	Parameter	Median	Lower 95%	Upper 95%	P(>0)
Biomass (Gaussian)	Symbiont status (S+)	-0.001	-0.436	0.436	0.498
Biomass (Gaussian)	Symbiont status (S+) × Precipitation	0.010	-0.390	0.428	0.517
Inflo count (NB)	Symbiont status (S+)	-0.449	-1.007	0.104	0.054
Inflo count (NB)	Symbiont status (S+) × Precipitation	0.185	-0.388	0.719	0.757
Total spikelets projected (NB)	Symbiont status (S+)	-0.158	-0.550	0.233	0.208
Total spikelets projected (NB)	Symbiont status (S+) × Precipitation	-0.130	-0.459	0.184	0.214

Table S11: Greenhouse experiment: posterior summaries of the endophyte main effect (S^+) and its interaction with soil-origin precipitation for each fitness component. For each model, the table reports the median estimate, 95% credible interval (Lower and Upper), and the posterior probability that the effect is greater than zero ($P(\beta > 0)$).

S.1.3 Supporting Figures

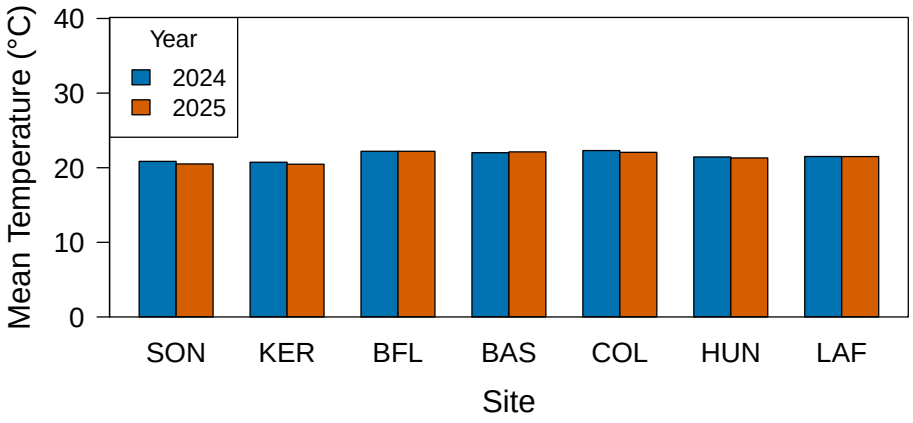


Fig S1: Annual temperature (°C) during the 2024 and 2025 census years. Sites are arranged west to east by longitude.

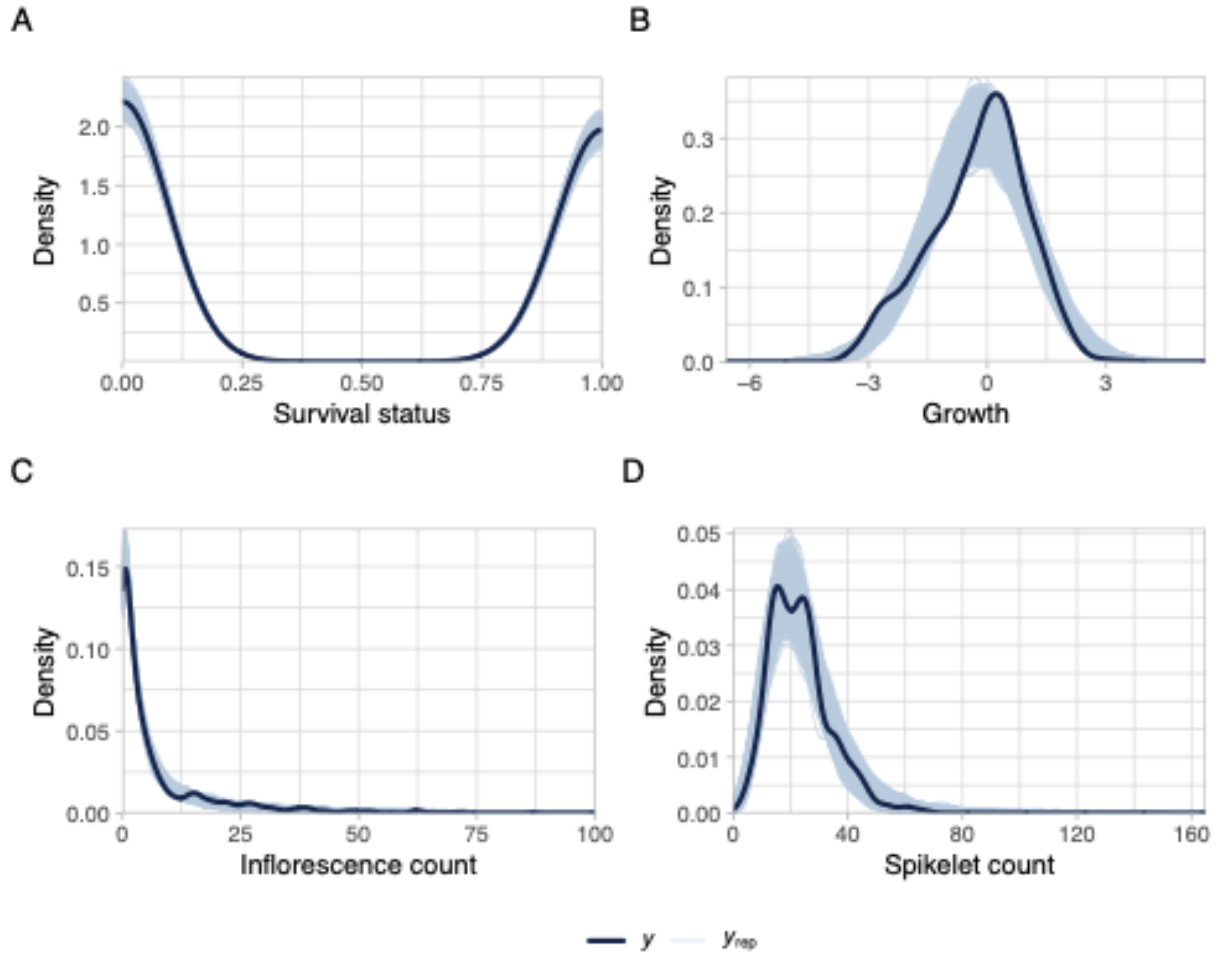


Fig S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y) and light blue lines show replicated datasets from posterior predictions (y_{rep}). Strong overlap across survival, growth, inflorescence, and spikelet production indicates adequate model fit for all vital rates.

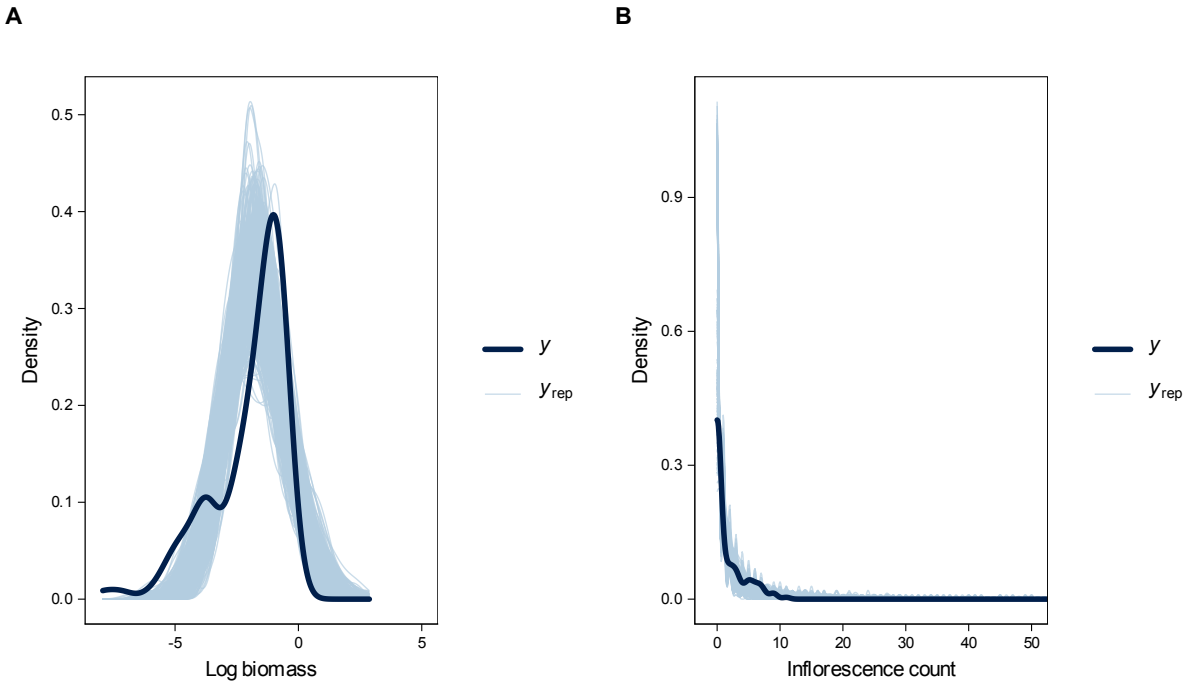


Fig S3: Posterior predictive checks for Bayesian models of plant performance in the greenhouse experiment. Density overlays show observed data (solid line) and 200 posterior predictive draws (shaded lines) for (A) aboveground biomass and (B) inflorescence count per plant. Models adequately capture the distribution of observed data across all response variables.

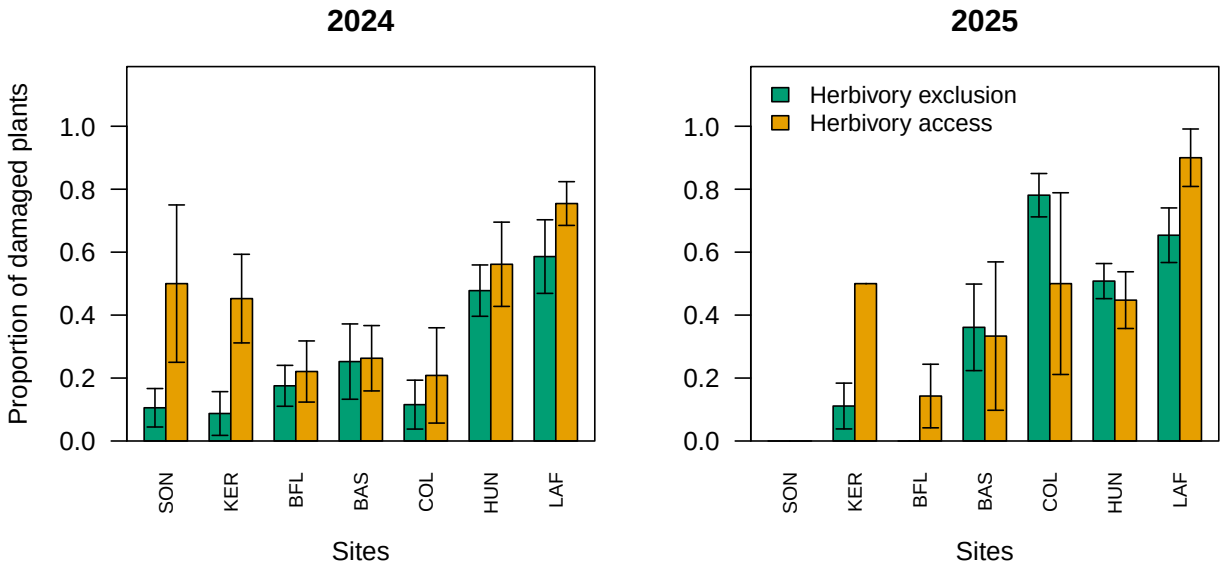


Fig S4: Proportion of plants damaged by herbivory across sites in 2024 and 2025. Sites are ordered west to east. Error bars indicate standard errors across plots within each site. Panels show results for 2024 (left) and 2025 (right).

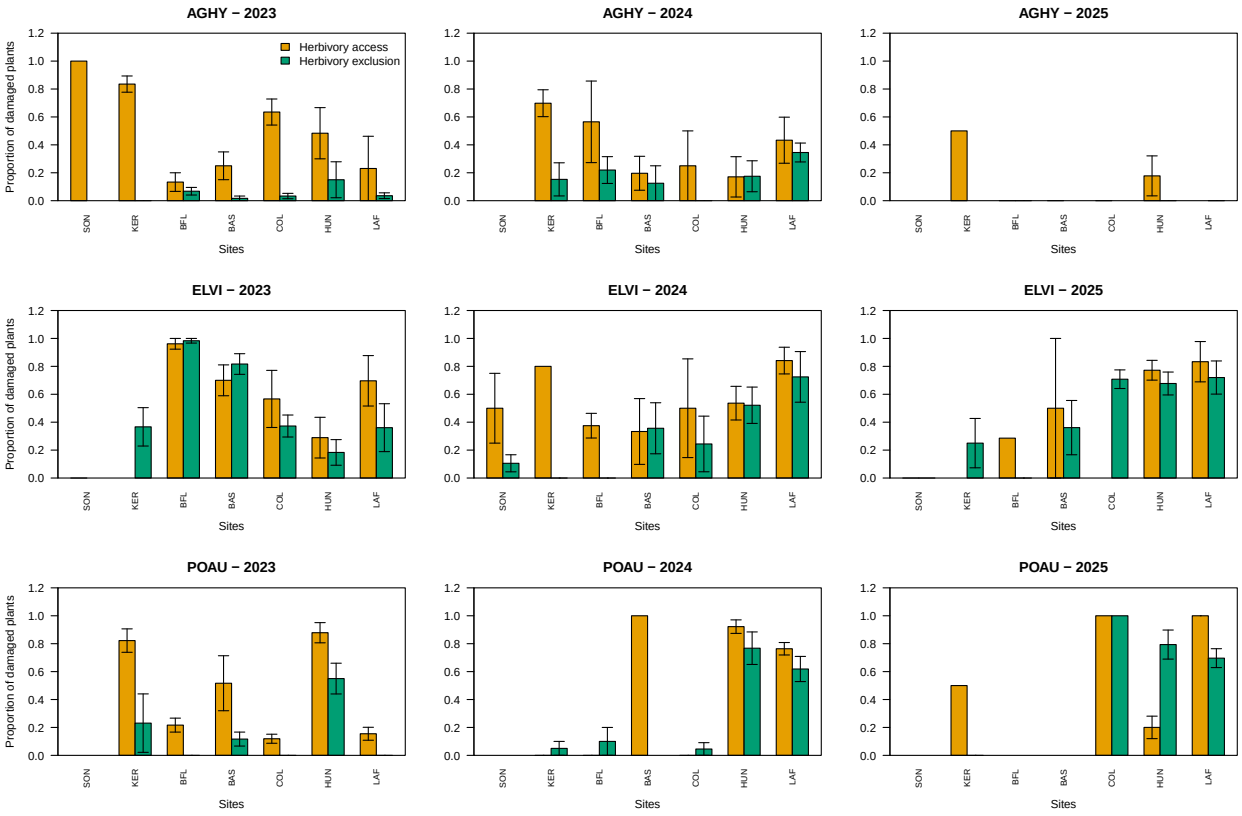


Fig S5: Proportion of plants damaged by herbivory across sites for *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* over 2023–2025. Sites are ordered west to east. Error bars indicate standard errors across plots within each site. Panels are arranged by species (rows) and year (columns).

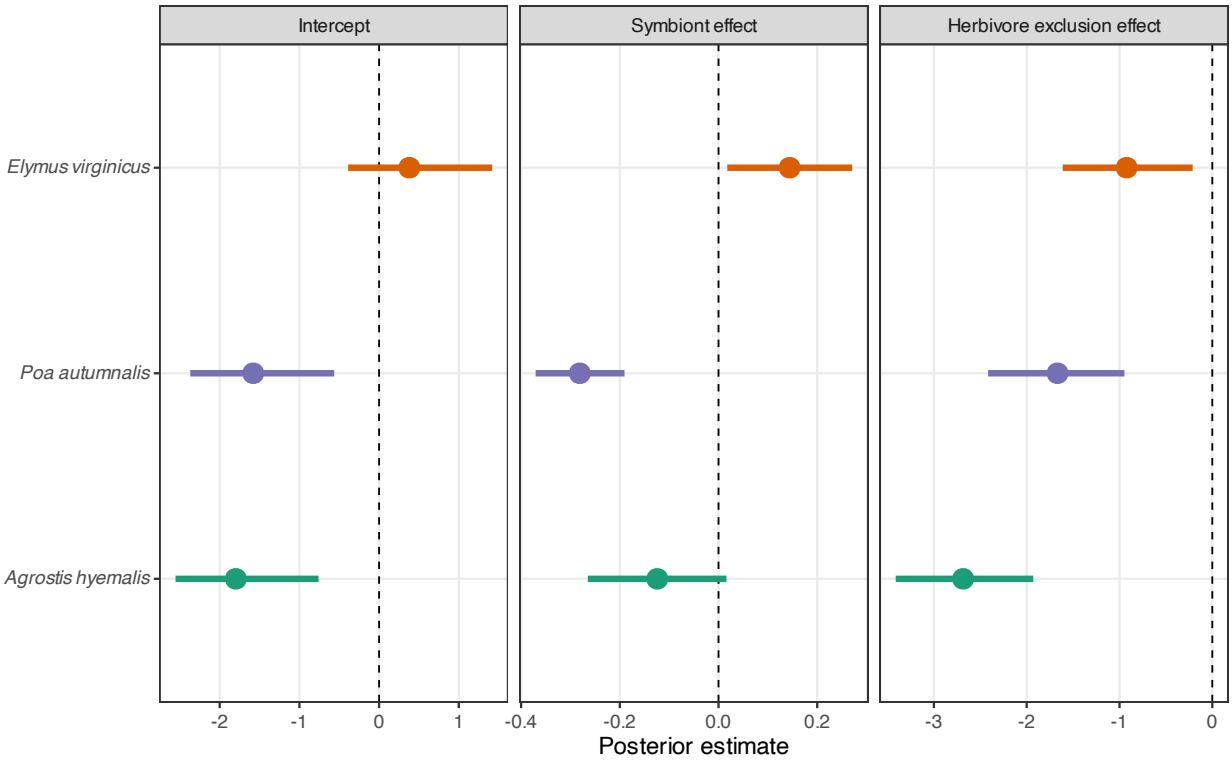


Fig S6: Posterior estimates (median and 90% credible intervals) of symbiont and herbivore exclusion effects on the proportion of damaged tillers for three host grass species, from a binomial model with site, year, and plot as random effects. Negative values indicate reduced herbivore damage, whereas positive values indicate increased damage. Back-transformed to predicted proportions of damaged tillers in herbivore-access plots, symbiosis reduced damage in *P. autumnalis* (S^+: 13.5% vs. S^-: 17.1%) and, more weakly, in *A. hyemalis* (S^+: 12.8% vs. S^-: 14.2%), and increased damage in *E. virginicus* (S^+: 62.8% vs. S^-: 59.4%). Points and intervals are colored by species: *Agrostis hyemalis* (green), *Elymus virginicus* (orange), and *Poa autumnalis* (purple). Dashed vertical lines indicate no effect.

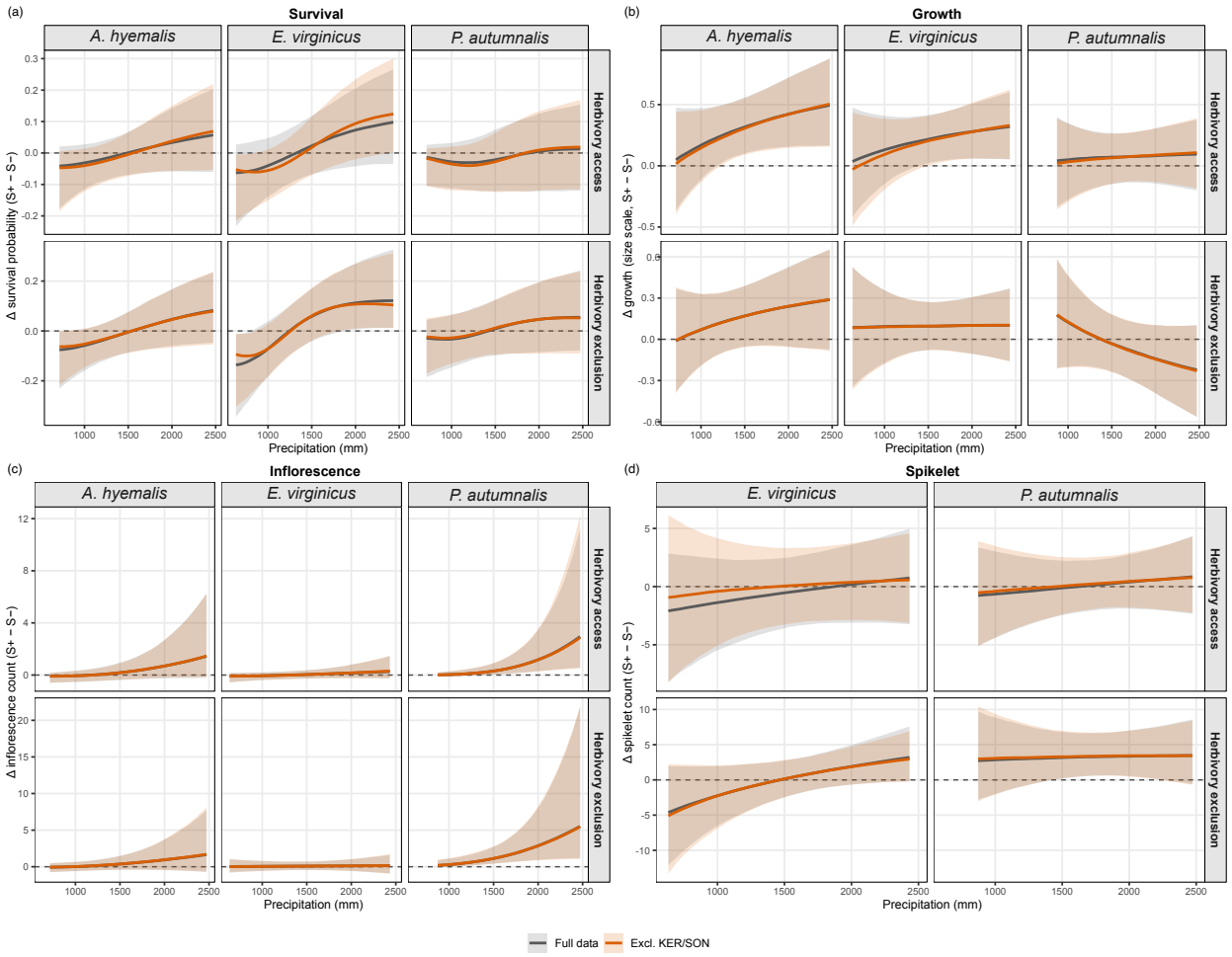


Fig S7: Sensitivity of estimated endophyte effects ($\Delta = S^{+} - S^{-}$) on (a) survival, (b) growth, (c) inflorescence production, and (d) spikelet production to exclusion of sites KER and SON, which were destroyed by deer and subsequently replanted in 2024. Lines and shaded bands show posterior median and 90% credible intervals for the full dataset (gray) and for the dataset excluding KER and SON (orange). Estimated effects were nearly identical between the two datasets across all species, vital rates, and herbivory treatments, indicating that the disturbance and replanting at these sites did not meaningfully influence our conclusions.

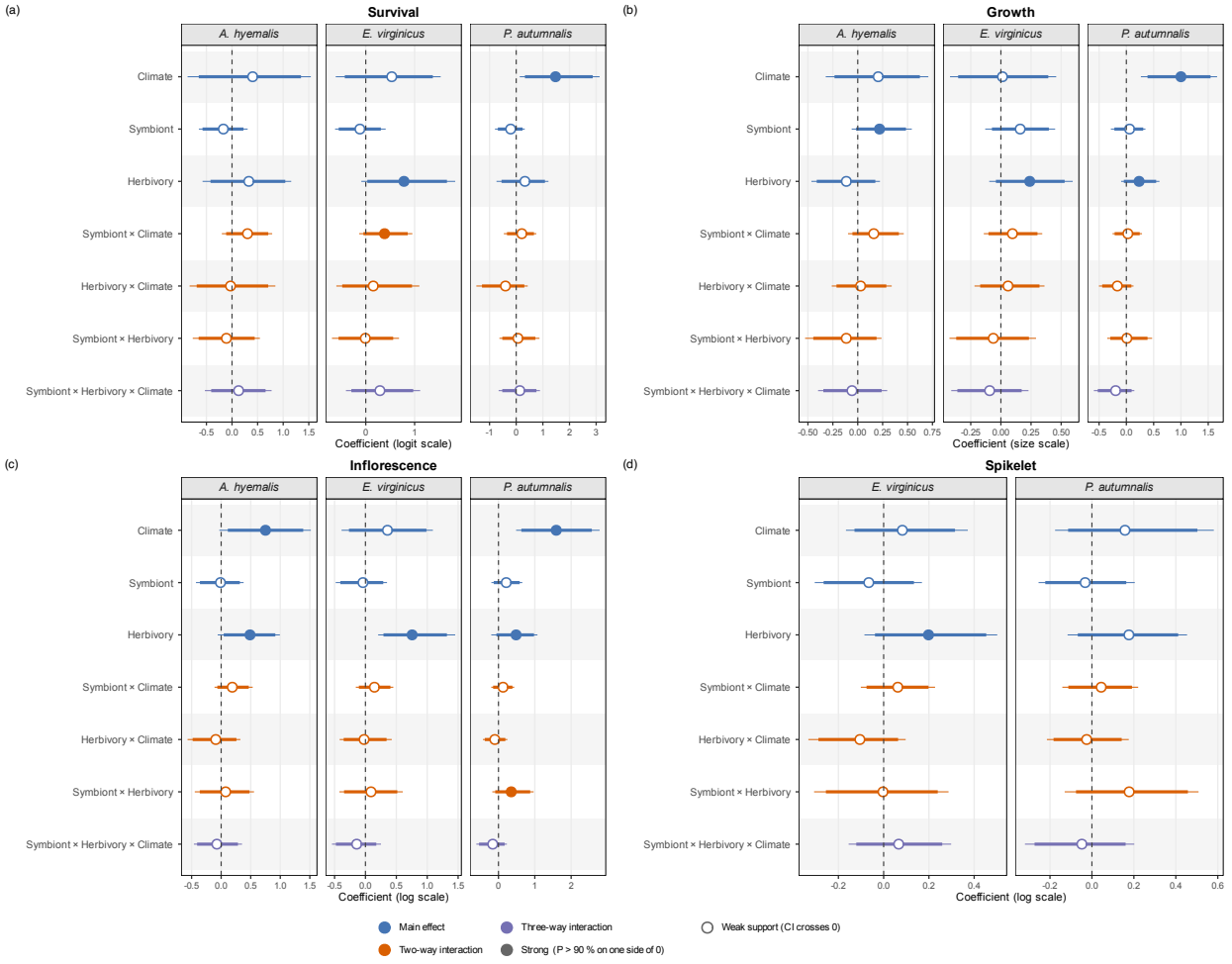


Fig S8: Posterior summaries of standardized regression coefficients from Bayesian models of survival, growth, inflorescence production, and spikelet production across three cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*; spikelet production was modeled for *E. virginicus* and *P. autumnalis* only). Panels show median posterior estimates (points), 90% credible intervals (thick bars), and 95% credible intervals (thin bars) for main effects of climate (precipitation), endophyte presence, and herbivory, as well as their two-way and three-way interactions. Filled points indicate strong posterior support for an effect ($\Pr(\Delta > 0)$ or $\Pr(\Delta < 0) > 0.90$), whereas open points indicate weaker support. Colors distinguish main effects, two-way interactions, and three-way interactions. Vertical dashed lines indicate no effect ($\beta = 0$). Coefficients are shown on the model link scale: logit for survival, linear size scale for growth, and log scale for inflorescence and spikelet production.

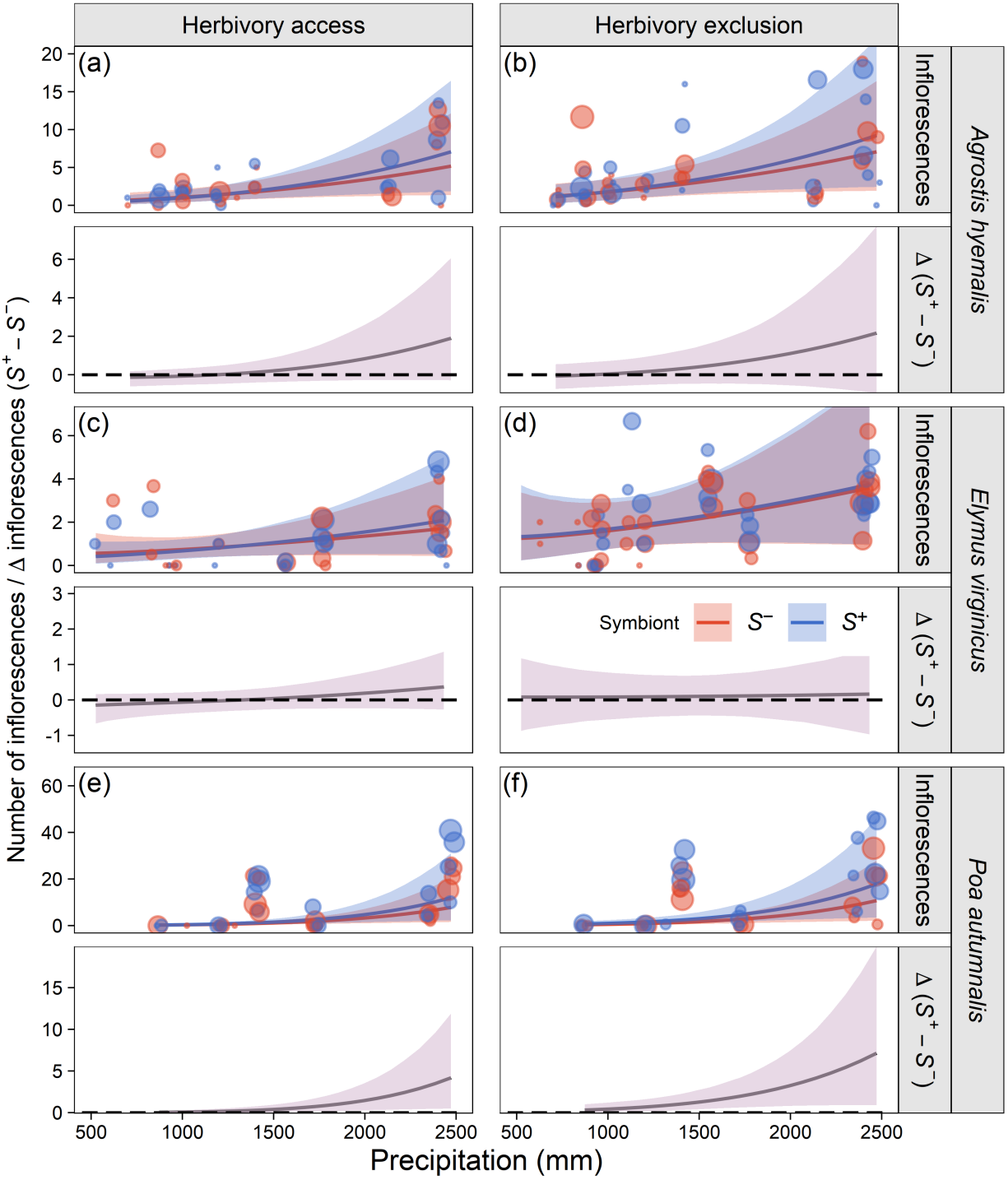


Fig S9: Predicted inflorescence production (upper panels) and symbiont effect ($\Delta = S^{+} - S^{-}$; lower panels) across the precipitation gradient. Lines show posterior means with 90% credible intervals. Points represent mean observed inflorescences per plot, slightly jittered for visualization. The dashed line at $\Delta = 0$ indicates no symbiont effect. Panels are arranged by species and herbivory treatment (herbivore access and exclusion). All panels share a common x-axis for illustrative purposes; some species extend beyond their observed precipitation range.

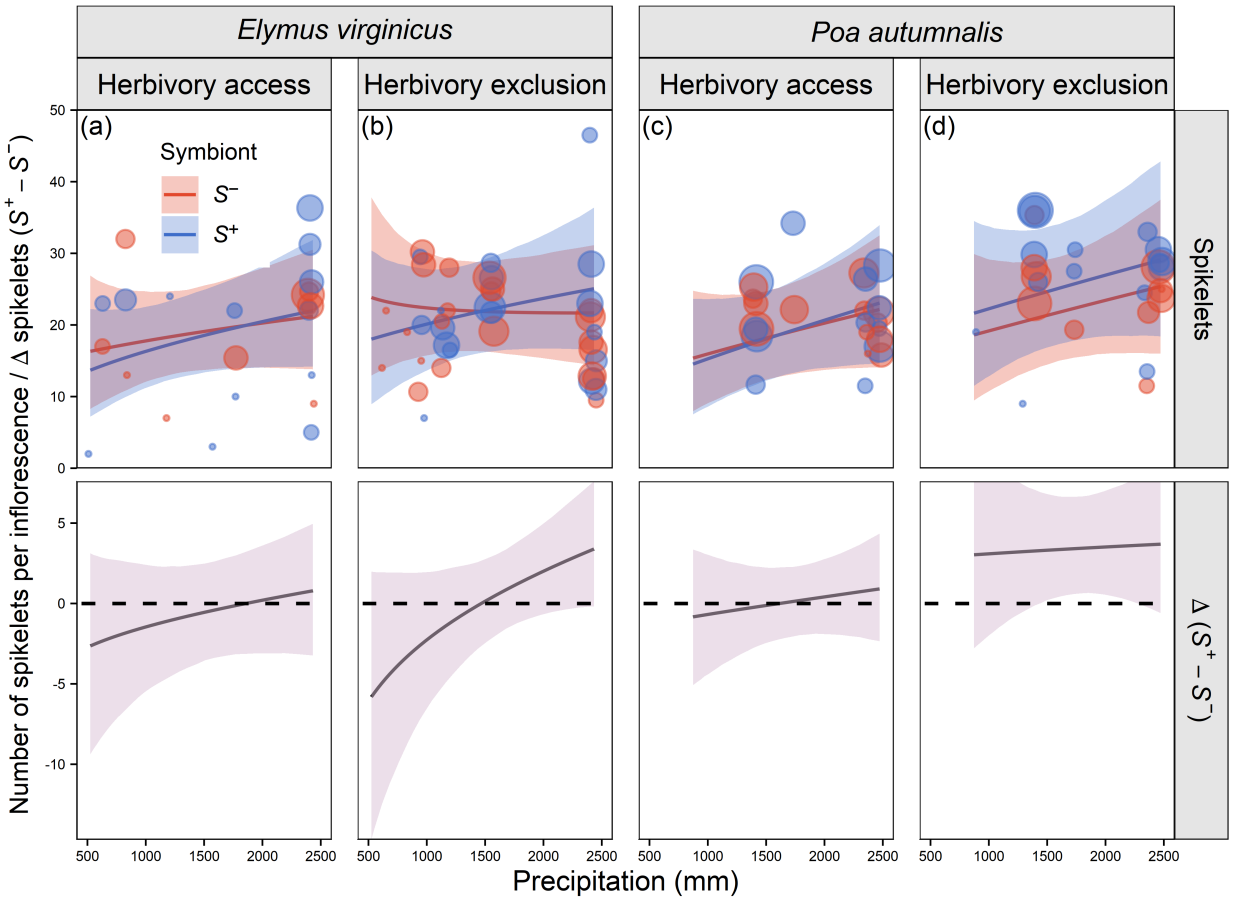


Fig S10: Predicted spikelet production (upper panels) and symbiont effect ($\Delta = S^{+} - S^{-}$; lower panels) across the precipitation gradient. Lines show posterior means with 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for two species only (see Methods). The dashed line at $\Delta = 0$ indicates no symbiont effect. Panels are arranged by species and herbivory treatment (herbivore access and exclusion). All panels share a common x-axis for illustrative purposes; some species extend beyond their observed precipitation range.

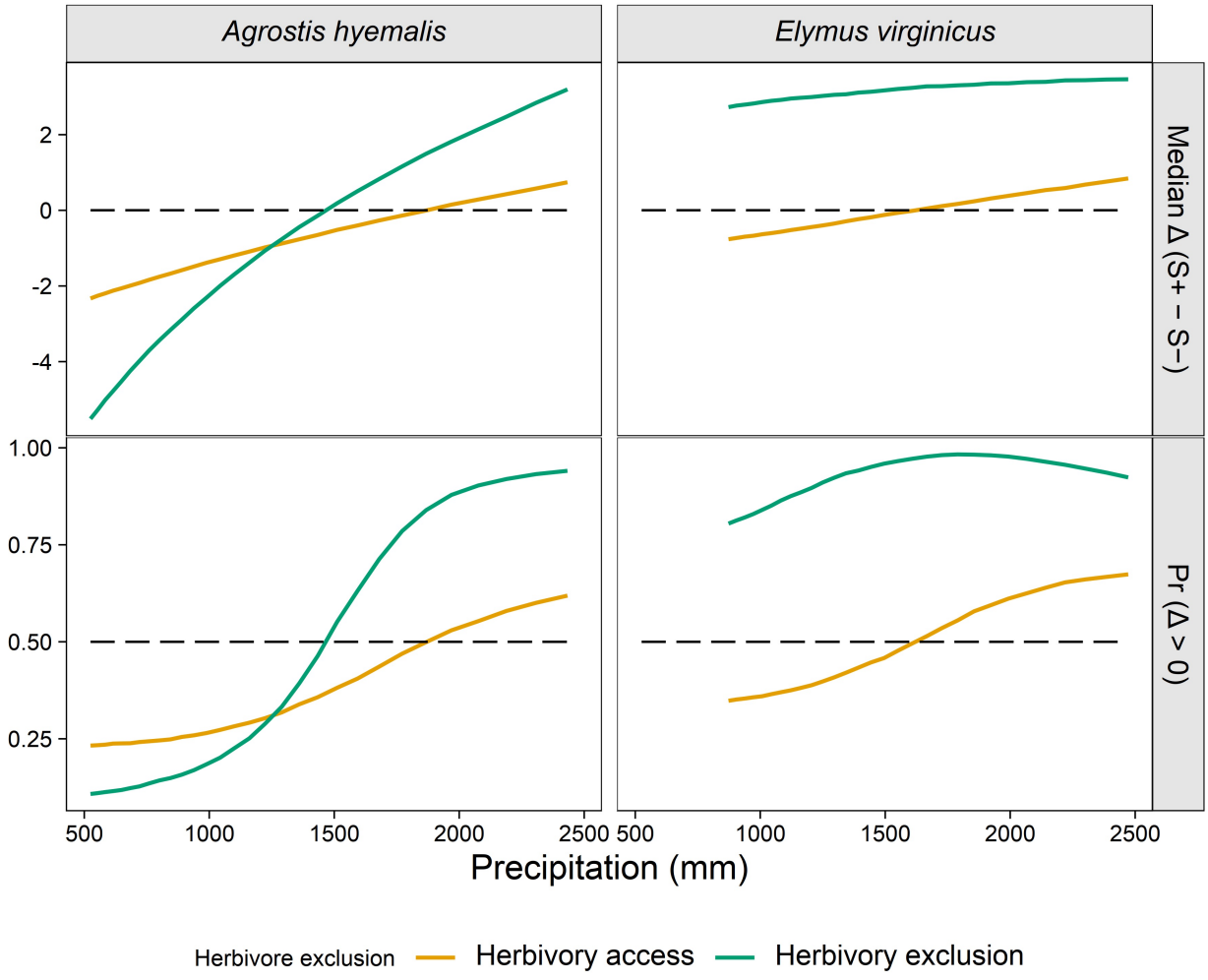


Fig S11: Median Δ $(S^+ - S^-)$ and posterior probability $P(\Delta > 0)$ for spikelet production across precipitation quantiles for each species and herbivory treatment. Herbivore exclusion treatments are indicated by colour. Horizontal dashed lines indicate $\Delta = 0$ for medians and $P(\Delta > 0) = 0.5$ for probabilities. Panels are arranged by metric and species.

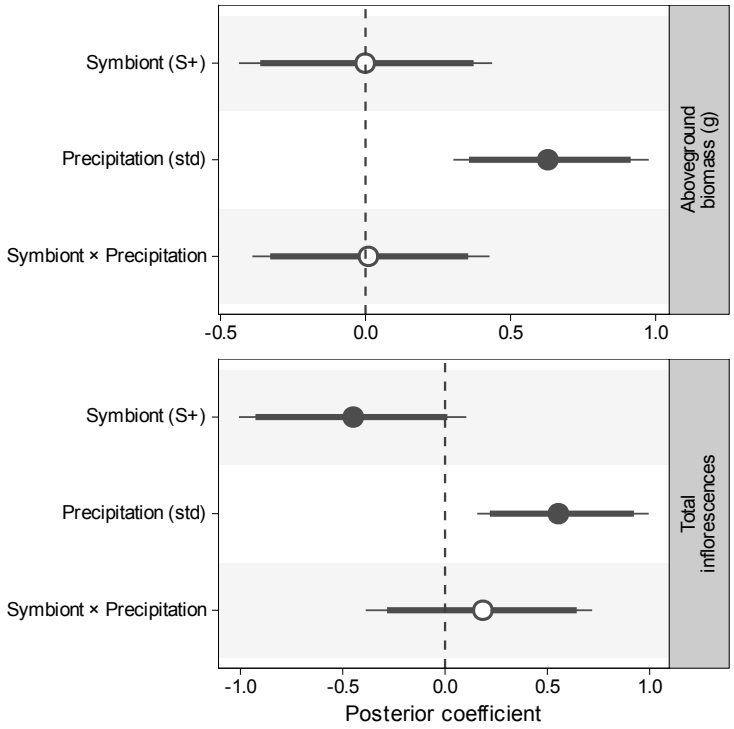


Fig S12: Posterior coefficient estimates for the greenhouse experiment (median, with thick and thin bars showing 90% and 95% credible intervals) for symbiont presence, soil-origin precipitation (standardized), and their interaction, for aboveground biomass (top) and total inflorescences (bottom). Filled points indicate estimates with posterior probability >0.9 of being above or below zero; the dashed line marks zero.

Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

Jacob K. Moutouama [*1, 2], Joshua C. Fowler[3], Akiem M. Gough[4], Julia Martin[4], Christopher C. Oxley[4], Karl Schrader[4], Karis Williams[4], and Tom E.X. Miller[4]

[1]Department of Botany, University of British Columbia, Vancouver, BC, Canada
[2]African Centre for DNA Barcoding, University of Johannesburg, Johannesburg, South Africa
[3]Department of Environmental Studies, University of Colorado Boulder, Boulder, CO, USA
[4]Department of BioSciences, Rice University, Houston, TX, USA

Running Title: Stress and Grass-Endophyte Symbioses
Keywords: demography, mutualism, plant-fungi interactions, range limits, stress gradient hypothesis, symbiosis
Article Type: Letter

Data accessibility statement: Upon acceptance, all data and computer scripts supporting the results will be archived in a Zenodo package. During peer review, all data files required to reproduce the analyses, including annual census data, are temporarily hosted via Dropbox and available at `https://www.dropbox.com/scl/fo/fgxng2mcemjyv41bihul9/AGiZs5mPRGcCZURpGyvhjCE?rlkey=7mgdyrus3ga44yj3ygfncfmkd&dl=1`. All analysis scripts (Stan and R) are available at `https://github.com/jmoutouama/endo-range-limits`.

Conflict of interest statement: None.

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This file contains: Abstract (150 words), Main Text (4998 words), Figures (1-6), References (73). Supporting Information is provided as a separate file.

*Corresponding author: jmoutouama@gmail.com

Abstract

Plant–microbe symbioses are expected to enhance host resilience to environmental stress, yet whether symbiont effects differ across abiotic versus biotic stress dimensions remains unclear. We tested how aridity and herbivory shape fitness effects of vertically transmitted fungal endophytes on three grass host species using herbivore exclusion experiments along a precipitation gradient in the south-central United States. Contrary to predictions, endophyte-associated hosts showed stronger declines in demographic performance with aridity than endophyte-free plants, suggesting abiotic stress exacerbated symbiont costs. Endophyte symbiosis reduced herbivory for two host species, consistent with protective benefits, but herbivore exclusion effects were weaker and more idiosyncratic relative to climate stress. A greenhouse experiment using soils from field sites showed endophyte effects were not strongly sensitive to soil origin, supporting a causal role of climate in driving site-level variation. Together, these results challenge the expectation that stress enhances symbiosis benefits and suggest limited buffering potential under increasing aridity.

Introduction

Mutualistic symbioses are widespread and ecologically important, often mediating stress tolerance and defense against natural enemies for host organisms. Host-symbiont interactions, like mutualisms more broadly, are strongly context-dependent, with outcomes shaped by environment (Bronstein, 1994; Hoang et al., 2024; Hoeksema and Bruna, 2015; Rudgers et al., 2020). The stress-gradient hypothesis (SGH) is a classic conceptual framework for variation in interaction outcomes, predicting that the frequency of positive interactions increases under stressful conditions (Bertness and Callaway, 1994; Holmgren and Scheffer, 2010; Louthan et al., 2015; Maestre et al., 2009). For example, microbial symbionts that are costly to maintain under benign conditions may benefit host plants under drought stress (Giauque et al., 2019). Stress can take many forms, including abiotic and biotic factors, and it remains unclear how predictions of the SGH for host-symbiont interactions play out over different dimensions of environmental stress.

Under biotic stress, such as competition or attack by natural enemies, symbionts can mediate defensive traits across diverse taxa (Atala et al., 2022; Bastias et al., 2017; Flórez et al., 2015; Haine, 2008; Vega, 2008). Similarly, under abiotic stress, including drought, nutrient limitation, and thermal extremes, symbiotic partners often enhance host tolerance through improved resource acquisition, physiological regulation, and stress-responsive metabolism (Afkhami and Rudgers, 2008; Cesar et al., 2024; Clay and Schardl, 2002). In plants specifically, microbial symbionts such as mycorrhizal fungi, rhizobia, and endophytic bacteria and fungi can promote water and nutrient uptake, nitrogen fixation, and resistance to herbivores, allowing hosts to maintain growth and function under harsh abiotic or biotic conditions (Amijee et al., 1989; Friesen et al., 2011; van Der Heijden et al., 2015). Many plant symbionts also stimulate the production of antioxidant enzymes, reducing oxidative damage caused by stressors such as drought, salinity, and heat (Ruiz-lazona et al., 1996). By improving resource acquisition, modulating physiology, and enhancing stress-responsive metabolism, these interactions help host plants buffer against environmental extremes (Fowler et al., 2023; Li et al., 2026; Rudgers et al., 2020).

Environmental stress can shift the balance of costs and benefits between hosts and their symbiotic microbes, and the direction and magnitude of this shift can depend on stress severity and environmental context. In plants, maintaining symbionts requires hosts to allocate valuable carbon and nutrients to their microbial partners; for example, mycorrhizal fungi can demand up to 20% of a host's photosynthetically fixed carbon (Soudzilovskaia et al., 2015; Thirkell et al., 2020). Under stress, this investment can outweigh benefits, reducing survival, growth, or reproduction (Bronstein, 2001; Johnson et al., 1997; Klironomos, 2003). A parallel dynamic occurs in coral bleaching, where heat stress can disrupt the coral-algal mutualism, with thermal thresholds for breakdown relatively well quantified compared with those in most

other symbioses (Evensen et al., 2022; Marangon et al., 2025; Vidal-Dupiol et al., 2009). Across systems, symbiotic microbes can therefore either mitigate or exacerbate host responses to environmental stress, depending on stress severity, the type and combination of stressors, and characteristics of the interacting partners (David et al., 2018; Kawai and Tokeshi, 2007; Le Roux and McGeoch, 2010; Michalet, 2007). Thus, although the classic SGH predicts that positive interactions should strengthen with increasing stress, extensions of the hypothesis recognize that positive interactions can weaken or become neutral or antagonistic under sufficiently severe stress. However, the environmental and biological conditions under which the balance shifts from beneficial to neutral or detrimental remain poorly understood. Identifying these conditions is essential for predicting when stress will strengthen mutualistic outcomes and when it will instead impose demographic costs on hosts.

Despite growing interest in symbiosis and host demography under environmental stress, few studies have tested how abiotic and biotic stressors synergistically shape the costs and benefits of symbiosis to host organisms. In a greenhouse study of *Ammophila breviligulata*, symbiont-associated plants grew more than symbiont-free plants under either herbivory or drought alone, but this benefit disappeared when the two stressors were combined, with symbiont-associated and symbiont-free plants showing similar growth (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors under natural conditions is particularly important in the context of global change (Louthan et al., 2015), as shifting climate patterns and altered herbivore pressures may change the balance of costs and benefits in plant–microbe interactions, potentially influencing species distributions and ecosystem function (Reynolds et al., 2003). Understanding how abiotic stress, biotic stress, and microbial symbiosis interact requires experiments that manipulate all three factors simultaneously, yet such fully factorial studies remain rare outside controlled settings. Here, we address this gap by examining the demographic effects of endophyte symbiosis on three grass host species spanning a natural range-core/range-edge precipitation gradient and in the presence/absence of herbivores.

Symbioses between cool-season grasses and seed-transmitted fungal endophytes (*Epichloë* spp.) provide a powerful experimental model for studying how abiotic and biotic elements of ecological context influence host–symbiont interactions (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen et al., 2010; Schardl et al., 2004). Because vertical transmission links host and symbiont fitness, it is expected to favor the evolution of mutualistic interactions (Gundel et al., 2011, 2024). Environmental conditions, including long-term variation in precipitation, temperature, and herbivore exposure, strongly influence both the prevalence of *Epichloë* infection and the benefits conferred to hosts (Clay et al., 2005; Fowler et al., 2025; Rudgers et al., 2016). For example, grasses hosting *Epichloë* endophytes often outperform endophyte-free individuals under water limitation, though the magnitude of these benefits depends on both host and symbiont identity

(Decunta et al., 2021). A key benefit of *Epichloë* symbiosis is the production of fungal alkaloids that deter herbivores or reduce herbivore performance, thereby mediating plant–herbivore interactions (Crawford et al., 2010; Saikkonen et al., 2010). Fungal alkaloids may also modify host osmotic balance in ways that promote water use efficiency (Decunta et al., 2021). Endophytes may impose costs on their hosts, including the carbon resources required to sustain the fungus and reduced growth or reproduction when fungal demands exceed host resource supply (Bennett and Groten, 2022). Finally, *Epichloë* endophyte effects can differ in both magnitude and direction among host vital rates, such that symbiosis may benefit one vital rate while having neutral or negative effects on another. For example, symbiosis may not improve survival under stress but can still enhance growth or reproduction (Rudgers et al., 2012; Yule et al., 2013).

We established experimental gardens of three species of native cool-season grasses with and without fungal endophytes across an aridity gradient in the south-central United States to investigate how symbiosis affects multiple host vital rates under varying abiotic and biotic stress. At each of seven sites along the aridity gradient, we manipulated vertebrate and invertebrate herbivory (allowing herbivore access versus exclusion) to test its interaction with abiotic stress. We asked whether endophyte symbiosis is beneficial for host grasses, whether benefits are amplified under abiotic and/or biotic stress as predicted by the stress-gradient hypothesis (SGH), and how these stressors interact to shape demographic outcomes. We evaluated four alternative hypotheses (Fig. 1):

1. Context-independent mutualism: Endophytes confer consistent positive effects on host demography across environmental conditions (Fig. 1a).
2. Herbivory-dependent effects: Endophyte benefits are greater under herbivore exposure than under herbivore exclusion, independent of the abiotic environment, consistent with a defensive role of fungal alkaloids (Fig. 1b).
3. Climate-dependent effects: Endophyte benefits decline with increasing precipitation, such that symbiont effects are more positive under arid than wet conditions, independent of herbivore exposure (Fig. 1c).
4. Herbivory- and climate-dependent effects: The effect of endophyte symbiosis depends on the joint influence of herbivory and precipitation, such that exposure to herbivory amplifies endophyte benefits under more arid conditions (Fig. 1d).

Finally, to supplement the geographically-distributed field experiment and isolate the causal role of climatic factors, we used a greenhouse experiment with one host species to assess the influence of edaphic factors that co-vary with precipitation across experimental sites.

Materials and methods

Study system

Our study focused on three cool-season perennial grass species native to woodland, savanna, and grassland habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (Poaceae, subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a short-lived perennial that germinates from autumn to late winter and, in our Texas study region, typically flowers between April and June. *E. virginicus* is a larger, longer-lived species that flowers throughout spring and summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers in late spring. *A. hyemalis* hosts the endophyte *Epichloë amarillans* (White Jr, 1994), *E. virginicus* hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* hosts *Epichloë* PauTG-1 or *E. poae* (Gundel et al., 2020). Previous studies of these associations have generally found beneficial effects on host performance, although effects can vary among host species, vital rates, and environmental conditions (Davitt et al., 2011; Gundel et al., 2020; Rudgers and Swafford, 2009; Yule et al., 2013). Like other *Epichloë* species, these endophytes are primarily transmitted vertically through seeds but some are additionally capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtmann et al., 2014). In our study, stromata were never observed on *E. virginicus* or *Poa autumnalis* and were rarely observed on *A. hyemalis* ($<5\%$ of individuals), suggesting a dominant role of vertical transmission. Herbivory in this system is primarily from deer, generalist insects (aphids, grasshoppers), and small mammalian grazers.

Source populations and symbiont removal

Plants used in our experiment were derived from 11 natural source populations located throughout the ranges of the focal species (Fig. 2, Table S1). Seeds from these source populations (collected from ca. 30 individuals per population) were either heat-treated to remove endophytes or left untreated (control), intended to produce lineages of symbiont-free (S^-) and symbiotic (S^+) plants from the same genetic background. For *E. virginicus* and *P. autumnalis*, heat-treated seeds were placed in a drying oven at 60°C for five days. For *A. hyemalis*, seeds were heat-treated in a water bath at 60°C for 12 min. For all but two populations, the S^- plants used in our experiment were reared through one or more generations following heat treatment, which minimized the confounding potential of direct physiological effects of heating seeds (Table S2). Seeds from control and heat-treated lineages were surface-sterilized with a 10% bleach solution (Sahu et al., 2022).

Seedlings were transferred to potting soil and maintained in the Rice University greenhouse at 18–29 °C and ambient photoperiod. Once plants were sufficiently large, they were vegetatively propagated to generate the required sample sizes, with clonal identities tracked across individuals. We refer to each independently derived source individual (originating from a single seed) as a "genotype" and consider all of its vegetatively propagated descendants to belong to the same genotype, such that all clones share the same genetic background of the host and (for S^+) symbiont. The experiment included 840, 840, and 720 plants of *A. hyemalis*, *E. virginicus*, and *P. autumnalis*, respectively, with half assigned to each endophyte status (S^+ and S^-), from three, four, and four source populations, respectively. The breakdown of these aggregate totals among source populations, genotypes, and clonal replicates per genotype is reported in Table S1. Endophyte status of all plants was verified prior to field transplant using microscopy and immunoblot assays (see Supplementary Methods).

Common garden experimental design and climate data

We established common garden experiments with S^+ and S^- hosts at seven sites along an aridity gradient from Louisiana to central Texas, US (Fig. 2, Table S3). From west to east, mean annual precipitation increases more than three-fold across these sites while mean annual temperature is relatively consistent, based on climate data collected during the same period as our demographic surveys (Fig. 2d, Fig. S1). For this reason, we focus on precipitation as the primary abiotic factor that varied across sites. This geographic gradient overlaps and exceeds the natural range edges of our focal species, ensuring exposure to realistic climatic and biotic conditions (Fig. 2a, b, c).

To characterize climate at these sites during the study period (2023–2025), we collected monthly temperature and precipitation data from PRISM at 800 m resolution (Hart and Bell, 2015). For each site, we calculated mean temperature (°C) and cumulative precipitation (mm) over each demographic interval, from the census month in year t through the census month in year $t+1$.

At each of seven sites along the gradient, we established 24 plots (1.5 m $\times$ 1.5 m), eight per host species (*P. autumnalis* was planted at six sites, excluding the westernmost site (SON) since it is well outside this species' natural range: Fig. 2c). Each plot was planted with 15 individuals, with 12, 9, 6, or 3 S^+ plants and the remainder S^-, corresponding to initial endophyte frequencies of 80%, 60%, 40%, or 20% S^+, respectively. This variation in initial endophyte prevalence was intended to track change in prevalence from different starting conditions and is not further considered here. Within each site, each plot was randomly assigned to either herbivore access or herbivore exclusion (Fig. 2e). The herbivory treatment

targeted both vertebrate and invertebrate herbivores: exclusion plots had deer fencing on four sides and were sprayed with Sevin insecticide (active ingredient zeta-cypermethrin) 2–3 times per growing season following the manufacturer's protocol; herbivore access plots were unsprayed and fenced on two sides to control for any fence effect.

Greenhouse-raised plants were planted into the field experiment in February–March 2023. For each species at each site, we planted a total of 60 S^+ and 60 S^- plants, half of those experiencing herbivore access and half experiencing herbivore exclusion. Clonal replicates were arranged such that all plots had the same genetic diversity (number of genotypes) for each species. Plots at the two westernmost sites (KER and SON) were destroyed by deer soon after planting in 2023 and subsequently replanted in 2024 with stronger fencing. Because demographic responses were calculated between consecutive censuses, observations from replanted plots were treated as new t–$t+1$ demographic intervals. We also repeated the analyses excluding KER and SON to assess whether the disturbance and subsequent replanting influenced our results.

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May, and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, survival was recorded as alive or dead and, for surviving plants, size was measured as the number of tillers. For surviving plants, growth was quantified as the log ratio of tiller count between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$). We recorded the number of inflorescences per plant for all species and, for reproducing individuals of *Poa autumnalis* and *Elymus virginicus*, counted spikelets on a maximum of three inflorescences per plant. Spikelet counts were not recorded for *Agrostis hyemalis* because its inflorescences contain numerous, very small spikelets that are difficult to count reliably in the field.

We measured herbivore damage to assess the effects of symbiont status and herbivore exclusion treatment on realized herbivory. For each individual at each census, we counted the number of tillers with any signs of herbivory (leaf chewing or clipped tillers) and quantified the proportion of total tillers with herbivore damage. We also binned plants as damaged/undamaged based on the presence/absence of any damaged tillers, and tallied the proportion of damaged plants by plot, site, and herbivore treatment.

Statistical modeling

To assess how abiotic and biotic stress modify the demographic effects of endophyte symbiosis, we fit generalized linear mixed models for each vital rate response in a hierarchical Bayesian

statistical framework. All models shared the same hierarchical structure for their expected values and were applied to survival, growth, and fertility (inflorescence count and, for two species, spikelet count per inflorescence). Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, inflorescence count with a negative binomial hurdle model, and spikelet count with a negative binomial distribution. Hurdle and zero-inflated negative binomial models had nearly identical predictive performance (ΔELPD = 0.2, SE = 3.1). We retained the hurdle model because it distinguishes whether a plant flowered and, if so, how many inflorescences it produced. Its two components, a Bernoulli model for flowering and a zero-truncated negative binomial model for inflorescence count, both used the linear predictor in Eq. 1.

All models included species-specific fixed effects for symbiosis (S^+ or S^-), site- and year-specific precipitation, and herbivory (herbivore access/exclusion), as well as all two- and three-way interactions among them. Quadratic effects of precipitation provided minimal improvement over linear effects (LOOCV; Table S4), so we used the linear model, in which the expected value for observation i (μ_i) was:

$$\begin{aligned}\mu_i = {} & \beta_0[Sp_i] + \beta_P[Sp_i] * precip_i + \beta_E[Sp_i] * symb_i + \beta_H[Sp_i] * herb_i \\ & + \beta_{EP}[Sp_i] * symb_i * precip_i + \beta_{HP}[Sp_i] * herb_i * precip_i \\ & + \beta_{HE}[Sp_i] * herb_i * symb_i + \beta_{HEP}[Sp_i] symb_i * herb_i * precip_i \\ & + \phi_{\text{site}[i],Sp_i} + \omega_{\text{source}[i]} + \rho_{\text{plot}[i]} \qquad (1) \\ & \phi_{\text{site}[i],Sp_i} \sim N(0, \sigma^2_{site}) \\ & \rho_{\text{plot}[i]} \sim N(0, \sigma^2_{plot}) \\ & \omega_{\text{source}[i]} \sim N(0, \sigma^2_{source})\end{aligned}$$

Here, precip_i is the natural logarithm of cumulative precipitation (mm) over the demographic interval, standardized to mean zero and unit variance; symb_i indicates endophyte presence ($\text{symb}_i = 1$) or absence ($\text{symb}_i = 0$), and herb_i indicates fencing treatment: herbivore access ($\text{herb}_i = 0$) or exclusion ($\text{herb}_i = 1$). The model includes random effects accounting for background variation at multiple hierarchical levels: $\phi_{\text{site}[i],Sp_i}$ captures site-to-site heterogeneity that is not explained by precipitation, $\rho_{\text{plot}[i]}$ captures plot-to-plot variation within sites, and $\omega_{\text{source}[i]}$ captures variation associated with the provenance of transplants (the latter two are unique to species and therefore not indexed by species). Random effects for site, plot, and source are host species-specific but the variance parameters (σ_{site}, σ_{source}, σ_{plot}) are shared across species; this assumes, for example, that site-to-site heterogeneity is of a similar magnitude across species, even as the model allows for "better" and "worse" sites

to differ by species. This hierarchical structure allows the model to "borrow strength" across species (Gelman and Hill, 2007), improving estimates of fixed effects while still capturing species-specific responses to the environmental gradient.

Models were implemented in R (R Core Team, 2023) using Stan via `rstan` (Stan Development Team, 2024), with four chains of 3,000 iterations (1,000 warm-up), weakly informative normal priors on fixed effects, and diffuse priors on random-effect variances (full details in the accompanying code). Posterior predictive checks confirmed adequate model fit (Fig. S2).

We used posterior probability distributions of the parameter estimates to visualize predictions for how effects of symbiosis varied across abiotic and biotic contexts. For each herbivory treatment (Access or Exclusion), we computed posterior predictions for the vital rates of S^+ and S^- hosts across the precipitation gradient (Eq. 1). From these predictions, we calculated the difference for each posterior sample ($\Delta = \mu^+ - \mu^-$) to estimate median effects and 90% credible intervals. We then computed the posterior probability that $\Delta > 0$, which quantifies the probability that endophytes enhance performance, given the model and data. For instance, a high posterior probability (e.g., $\Pr(\Delta > 0) > 0.9$) indicates strong evidence of an endophyte benefit, whereas a low posterior probability (e.g., $\Pr(\Delta > 0) < 0.1$) indicates strong evidence of an endophyte cost. Intermediate probabilities (approximately 0.7–0.9 or 0.1–0.3) indicate moderate evidence for positive or negative effects, respectively, whereas probabilities near 0.5 indicate no evidence for a directional effect.

Greenhouse experiment

We used site variation to capture an aridity gradient, but other abiotic factors, especially edaphic conditions, covary across sites. In particular, the two westernmost sites (KER and SON; Fig. 2) are not only the driest but also occur on the Edwards Plateau of central Texas, with distinct limestone soils. If edaphic variation rather than precipitation drives the symbiont $\times$ precipitation interaction observed in the field, we would expect the interaction to persist in the greenhouse, where soils from contrasting sites are brought into a common climate. Conversely, if the field pattern reflects a genuine climate-mediated effect of the symbiosis, the interaction should disappear when plants are decoupled from the climate of their soil's origin. To distinguish between these alternatives, we conducted a greenhouse experiment using *A. hyemalis* and soils collected from each field site, crossing symbiont status (S^+/S^-) with soil origin (seven sites; Fig. 2d; Table S1; Supplementary Methods S.1.2). Using data on biomass and inflorescence production in the greenhouse, we fitted similar hierarchical Bayesian models as the field analysis, with symbiont status, soil-origin precipitation (30-year normal, standardised), and their interaction as fixed effects (Supplementary Methods S.1.2.3).

Posterior predictive checks showed good agreement between the observed data and data simulated from the posterior distributions, indicating that the models adequately captured the main features of the observed biomass and inflorescence production data (Fig. S3).

Results

Efficacy of aridity and herbivory treatments

As intended, realized precipitation and herbivory treatments tracked the abiotic and biotic gradients targeted by our design. Realized precipitation at the seven experimental sites corresponded well to 30-year climate normals, with wetter eastern sites and drier western sites (Fig. 2d). Herbivore access also increased realized herbivory: plants in herbivore-access plots experienced 24% tiller damage on average, compared to 14% in herbivore-exclusion plots (Fig. S4; Fig. S5; Fig. S6). Endophyte symbiosis reduced the predicted proportion of damaged tillers in *P. autumnalis* (S^+: 13.5% vs. S^-: 17.1%) and, more weakly, in *A. hyemalis* (S^+: 12.8% vs. S^-: 14.2%), whereas it increased damage in *E. virginicus* (S^+: 62.8% vs. S^-: 59.4%; Fig. S6).

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Annual survival was greater overall and more positively responsive to precipitation for the longer-lived hosts *E. virginicus* and *P. autumnalis* compared to the shorter-lived *A. hyemalis* (Fig. 3). Endophyte effects on survival depended most strongly on precipitation in *E. virginicus*. Under herbivore exclusion, effects of endophytes on survival shifted from strongly negative under dry conditions ($\Delta = -0.13$, $\Pr(\Delta > 0) = 0.02$) to positive under wetter conditions ($\Delta = 0.12$, $\Pr(\Delta > 0) = 0.97$). Under herbivore access the same directional shift occurred but with weaker magnitude ($\Delta = -0.06$ to 0.10, $\Pr(\Delta > 0) = 0.10$–0.89). A similar but weaker directional shift occurred in *A. hyemalis* under both herbivory treatments. For the host *P. autumnalis*, survival was strongly responsive to climate variation but there was no evidence that symbionts modified the response to climate stress, and there was little difference between herbivory treatments (Fig. 5, Fig. 3).

For growth, endophyte benefits increased with precipitation in *A. hyemalis* and *E. virginicus* under herbivore access, with strong positive effects of symbiosis under the wettest conditions (*A. hyemalis*: $\Delta = 0.49$, $\Pr(\Delta > 0) = 0.99$; *E. virginicus*: $\Delta = 0.32$, $\Pr(\Delta > 0) = 0.97$; Fig. 4). Under herbivore exclusion, *A. hyemalis* showed a similar positive trend but with weaker support ($\Delta = 0.29$, $\Pr(\Delta > 0) = 0.90$), while *E. virginicus* showed little variation across

the gradient ($\Delta = 0.09$–0.10, $\Pr(\Delta > 0) = 0.60$–0.75). For both *A. hyemalis* and *E. virginicus*, endophyte effects on growth were most beneficial under wetter abiotic conditions and herbivore access (Fig. 4a–d). As for survival, *P. autumnalis* growth rates declined strongly under drier conditions but endophyte effects were weak and highly uncertain (Fig. 5). Under herbivore exclusion, endophyte effects on *P. autumnalis* growth shifted from weakly positive to negative with increasing precipitation ($\Delta = -0.22$, $\Pr(\Delta > 0) = 0.13$), the only species / vital rate pair that was even directionally consistent with drought-protective benefits of symbiosis.

For inflorescence production, endophyte effects were strongly positive in *P. autumnalis* and increased with precipitation under both herbivory treatments (Fig. S9). Under herbivore access, effects ranged from weak under drier conditions ($\Delta = 0.02$, $\Pr(\Delta > 0) = 0.68$) to strongly positive under wetter conditions ($\Delta = 2.91$, $\Pr(\Delta > 0) = 0.99$). Under herbivore exclusion, benefits were consistently strong across the gradient ($\Delta = 0.19$–4.97, $\Pr(\Delta > 0) = 0.97$–1). In contrast, endophyte effects on inflorescence production were weaker and highly uncertain across the gradient in the hosts *A. hyemalis* and *E. virginicus* under both herbivory treatments (Fig. S9; Fig. 5). Spikelet production in *E. virginicus* and *P. autumnalis* was climate-sensitive; endophyte effects on spikelets increased with precipitation under herbivore exclusion but were weak under herbivore access (Fig. S10; Table S10).

Figure 5 synthesizes statistical confidence in endophyte effects across host species, vital rates, and abiotic and biotic stress gradients. The broad trend that emerged was a shift from negative or neutral endophyte effects under drier conditions to more positive effects under wetter conditions. These patterns were largely consistent with the hypothesis that abiotic (climate) stress is the dominant modifier of endophyte effects (Hypothesis 3; Fig. 1c), but in the opposite direction from that predicted, as endophyte benefits tended to increase rather than decrease with precipitation. Partial support emerged for herbivory and combined effects of herbivory and climate as symbiosis modifiers (Hypotheses 2 and 4; Fig. 1b,d), depending on species and fitness component. Herbivory played a secondary role for most species and vital rates. Herbivore exclusion generally dampened the growth benefits that hosts gained from endophytes, consistent with an anti-herbivore protective role of *Epichloë* endophytes. However, the role of herbivory as a modifier of endophyte effects on other vital rates was more idiosyncratic.

Plots at the two westernmost sites (KER and SON) were destroyed by deer and replanted in 2024. We therefore repeated all analyses excluding these sites to assess whether their replanting influenced our conclusions. Excluding KER and SON produced qualitatively similar patterns of symbiont effects across climate and herbivory treatments (Fig. S7, Table S6), indicating that our conclusions were not driven by the replanted sites.

Greenhouse experiment: soil legacy effects do not mediate endophyte responses

In the greenhouse experiment with *A. hyemalis* growing on soils from each experimental site, fitness components responded strongly to soil origin even under uniform conditions: plants grown in soils from wetter sites produced greater aboveground biomass and more inflorescences than those from drier-origin soils (Figs. 6a–b), confirming that soil properties co-vary with the field precipitation gradient. However, the estimated symbiont effects on both aboveground biomass and inflorescence production showed little systematic variation with soil-origin precipitation (Fig. S12). Thus, although soil origin affected plant performance, it did not explain the variation in symbiont effects associated with precipitation in the field.

Discussion

Drawing on three years of field experiments and Bayesian analysis of host demography, we evaluated how biotic and abiotic stressors jointly influence the demographic effects of vertically transmitted *Epichloë* endophytes on three cool-season perennial grasses across a natural climatic gradient. We predicted that endophytes would enhance host fitness under aridity and/or herbivore pressure (Fig. 1), in line with stress amelioration and defense scenarios. Our results are inconsistent with context-independent mutualism (Fig. 1a): endophyte effects varied substantially across host species and along the precipitation gradient rather than remaining constant. They are also inconsistent with the climate-dependent hypothesis as formulated (Fig. 1c), which predicted more positive endophyte effects under arid than wet conditions; instead, benefits were strongest under benign, mesic conditions and weakened as abiotic stress increased, the opposite of the predicted direction. Support for the herbivory-dependent hypothesis (Fig. 1b) was mixed: effects of herbivore exclusion on endophyte benefits were inconsistent across species and fitness components, although the strongest effects were in the predicted direction, with herbivore exclusion dampening endophyte benefits for growth. We found little evidence for the herbivory–climate-dependent hypothesis (Fig. 1d), as herbivory did not consistently amplify endophyte benefits under more arid conditions. More broadly, the benefits of symbiosis depended on stress severity and host performance: they were greatest when symbiont-free hosts performed relatively well, but diminished or turned negative when those hosts performed poorly. This pattern is consistent with extensions of the SGH in which positive interactions weaken or shift toward antagonism under sufficiently severe stress (Kawai and Tokeshi, 2007; Le Roux and McGeoch, 2010; Michalet, 2007).

The physiological mechanisms underlying the decline in endophyte benefits with increasing aridity are unclear, but several non-exclusive processes may contribute. Reduced water availability may limit the host's capacity to benefit from the symbiosis through constrained nutrient uptake, reduced carbon gain, or drought-induced hormonal shifts that impair endophyte function or alter the cost–benefit balance of the interaction (Ahlholm et al., 2002; Bastías et al., 2024; Bronstein, 2001; Cui et al., 2022; Faeth, 2002). This interpretation aligns with the microbial mitigation–exacerbation continuum (David et al., 2018), which predicts that symbionts can shift from beneficial to costly as environmental stress exceeds the host's capacity to sustain the interaction. Our results are also anticipated by empirical patterns along precipitation gradients in other grass–*Epichloë* systems (Rudgers and Swafford, 2009), and by a systematic review noting that positive endophyte effects on drought tolerance are inconsistent across species and understudied for fitness measures such as fecundity and survival (Decunta et al., 2021).

A structural feature of the symbiosis may also contribute to this pattern. Within a host, endophyte infections last for the life of the infected tissue and are lost only at reproductive transitions, from tillers to seeds or seeds to seedlings (Afkhami and Rudgers, 2008). Hosts therefore cannot clear a costly symbiont from already-infected tissue within a single generation, even as conditions deteriorate. This may create a mismatch between environmental change and symbiont turnover, allowing populations to temporarily retain associations that no longer provide demographic benefits. We did not assess changes in endophyte prevalence during the experiment and therefore cannot test this possibility directly. The consequences of this constraint may depend on the degree of host–symbiont specialization and on functional differences among endophytes, which can alter both the costs of maintaining the symbiosis and its capacity to confer benefits under stress (Saona et al., 2010). Testing these mechanisms will require comparing non-structural carbohydrates and tissue nitrogen between plants with and without symbionts across aridity gradients and among host–endophyte systems, to determine whether changes in host resource status help explain why endophyte benefits strengthen under stress in some systems but weaken in others.

Host species differed in how strongly increasing aridity altered the demographic benefits of *Epichloë*. *P. autumnalis* maintained strong reproductive benefits across the climatic gradient, whereas effects on survival and growth were weaker. In contrast, *A. hyemalis* and *E. virginicus* showed more pronounced climate-dependent responses in survival and growth, suggesting that these vital rates may be more responsive to climatic variation near their arid range margins. These contrasts may reflect differences in host physiological tolerance, symbiont identity, or the ecological contexts in which each species evolved (Decunta et al., 2021), and

highlight the importance of considering multiple vital rates when evaluating grass–endophyte symbioses under climate stress (Cheplick and Faeth, 2009).

Herbivore exclusion generally dampened endophyte benefits for growth across species, consistent with an anti-herbivore protective role of *Epichloë* endophytes reinforced under natural herbivore pressure (Clay, 1988; Decunta et al., 2021). Effects of herbivory on survival and reproduction were more idiosyncratic, varying across species and precipitation contexts, suggesting these consequences are not uniformly mediated by herbivore pressure across fitness components. Endophytes reduced herbivore damage in *A. hyemalis* and *P. autumnalis*, in keeping with alkaloid-mediated defense, but increased damage in *E. virginicus* (Fig. S6), possibly because symbiotic plants were more vigorous and therefore more attractive to herbivores. These species-specific responses may reflect differences in alkaloid identity and potency among endophytes (Siegel et al., 1990), together with host condition. Thus, the effects of endophytes on herbivory may depend on both the defensive chemistry of the symbiont and the condition of the host.

The greenhouse experiment suggests that the symbiont–climate interaction observed in the field is more closely linked to the direct physiological effects of water availability on the host than to soil properties that co-vary with aridity. Although soil origin affected plant performance under common watering conditions, the negligible symbiont $\times$ soil-origin precipitation interactions for biomass and inflorescence production indicate that this soil gradient did not reproduce the climate dependence of symbiont effects observed in the field. Thus, soil properties can influence host performance, but are unlikely to be the primary driver of climatic variation in symbiont benefits. Instead, direct effects of water availability on host carbon balance, growth, and resource allocation may influence the strength of the association.

The precipitation gradient spanned by our common gardens overlaps with the range margins of these species, making our results relevant to a further question about mutualism and species range dynamics. As environmental change reshapes endophyte prevalence in grasses (Fowler et al., 2025; Li et al., 2026), it remains unclear whether symbiosis can buffer host populations at these margins or instead becomes less effective as conditions become more arid. Future work should determine whether climate change affects grass–endophyte associations primarily by altering endophyte persistence itself, or by weakening the benefits endophytes provide to hosts at range margins, since these two mechanisms imply different trajectories for populations at the edge of the species' range.

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Figures

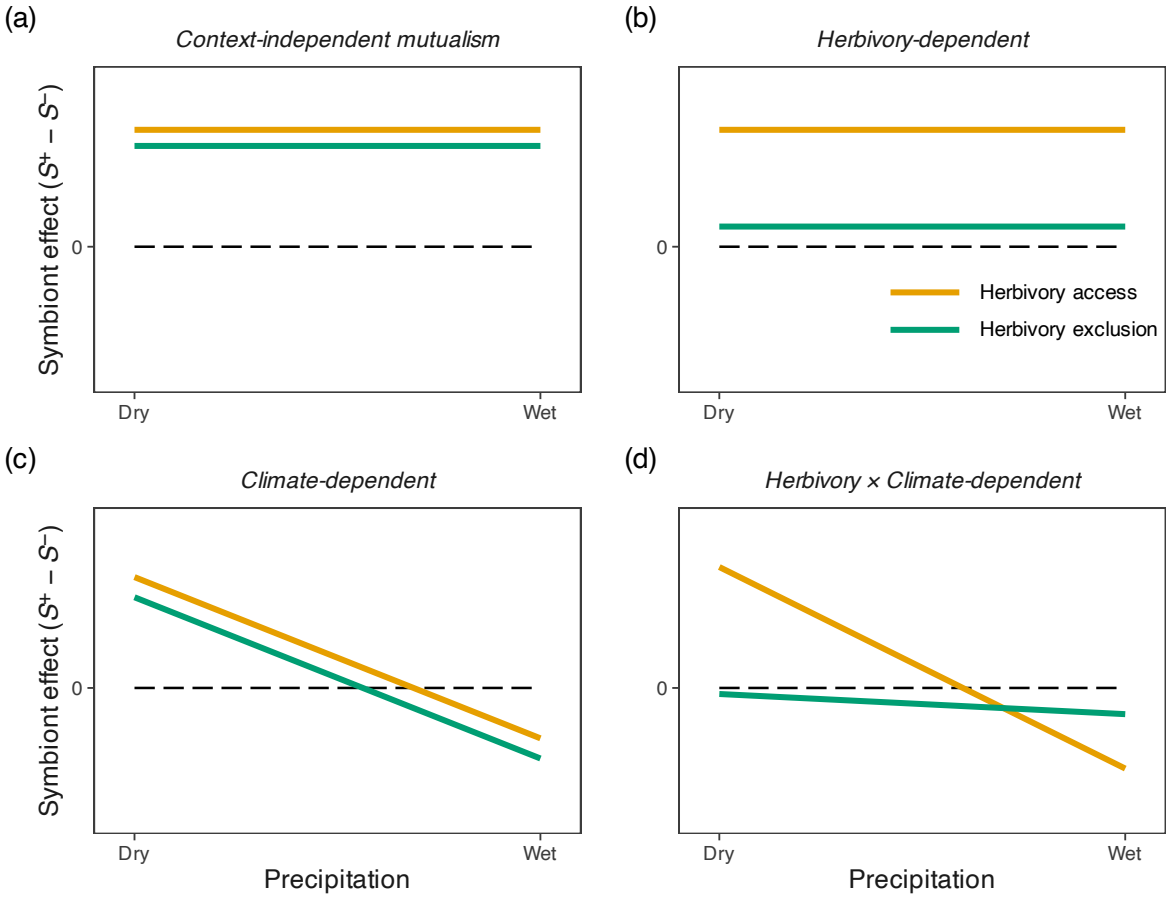


Figure 1: Competing hypotheses for how plant–symbiont interactions vary along a precipitation gradient under contrasting herbivory regimes. Positive values for the demographic contrast of $S^+ - S^-$ indicate beneficial effects, and negative values indicate costly effects of symbionts. (a) Context-independent mutualism, where symbiont effects remain constant across environmental conditions and herbivory treatments. (b) Herbivory-dependent interactions, where the effect of symbionts differs in magnitude between herbivory access and exclusion but is insensitive to climate. (c) Climate-dependent interactions, where symbiont effects shift systematically along the precipitation gradient, independent of herbivory. (d) Combined herbivory- and climate-dependent interactions, where both environmental context and biotic pressure jointly shape symbiont effects, resulting in non-parallel and potentially decoupled response trajectories. Together, these scenarios illustrate alternative hypotheses for how symbiotic interaction outcomes may vary across abiotic and biotic stress gradients.

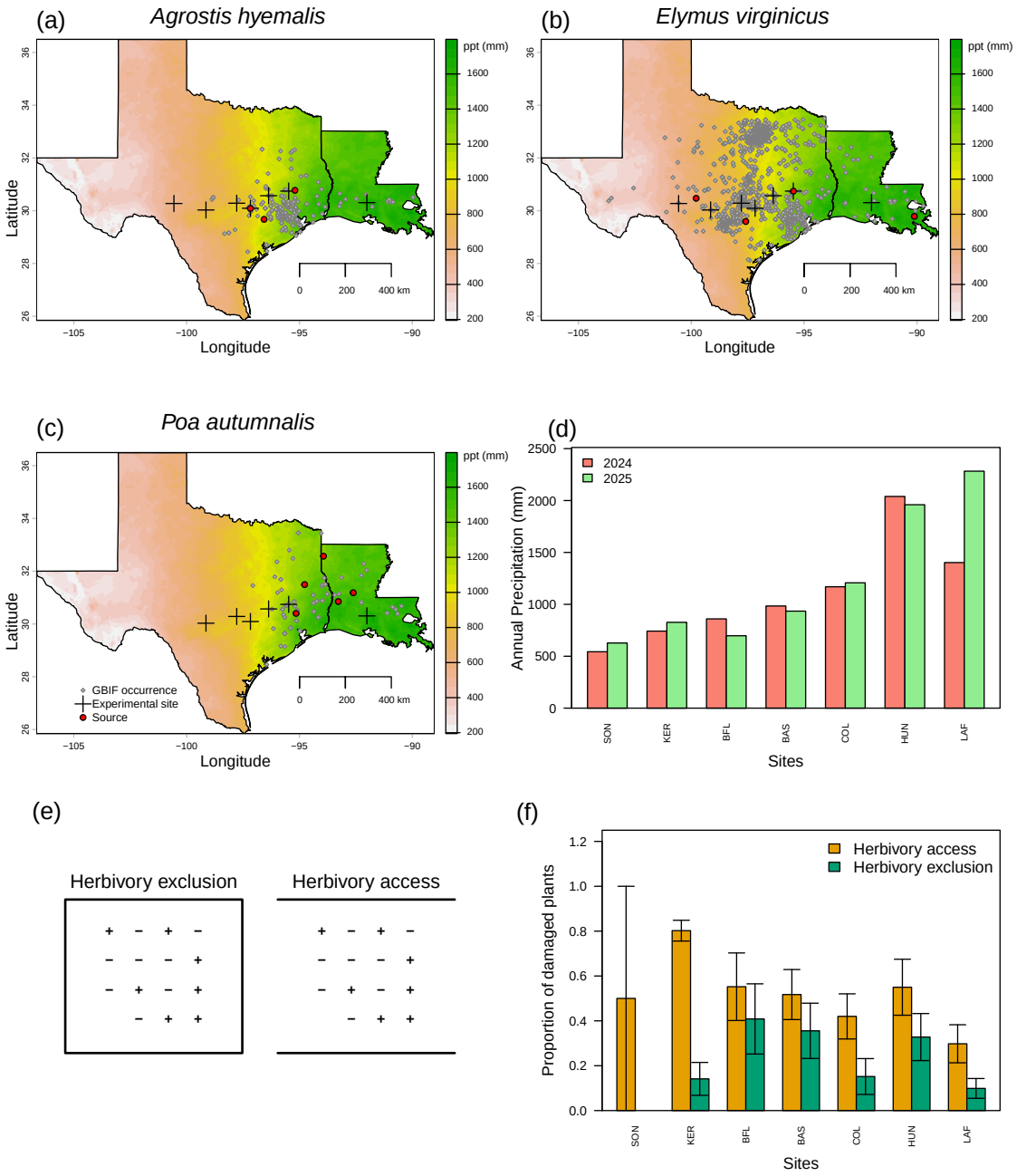


Figure 2: Geography of common garden sites along a natural aridity gradient from Texas to Louisiana, USA for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals for average annual precipitation (1996–2025) from PRISM (Hart and Bell, 2015), illustrating the precipitation gradient across experimental sites (scale bar in mm). Red dots indicate source population locations; grey dots represent GBIF occurrence records within the study area (GBIF, 2025a,b,c). (D) Cumulative precipitation over the demographic intervals ending in 2024 and 2025 at each experimental site (ordered west to east), averaged across species; derived from PRISM climate data. (E) Schematic illustration of herbivore exclusion and herbivore access plots; '+' denotes symbiont-positive and '-' denotes symbiont-negative individuals. Equal numbers are shown for illustration, although symbiont frequencies varied among experimental plots. (F) Proportion of damaged plants in herbivore exclusion and herbivore access plots across experimental sites (mean ± SE). Species-specific breakdowns are provided in Supplementary Figs. S4 and S5.

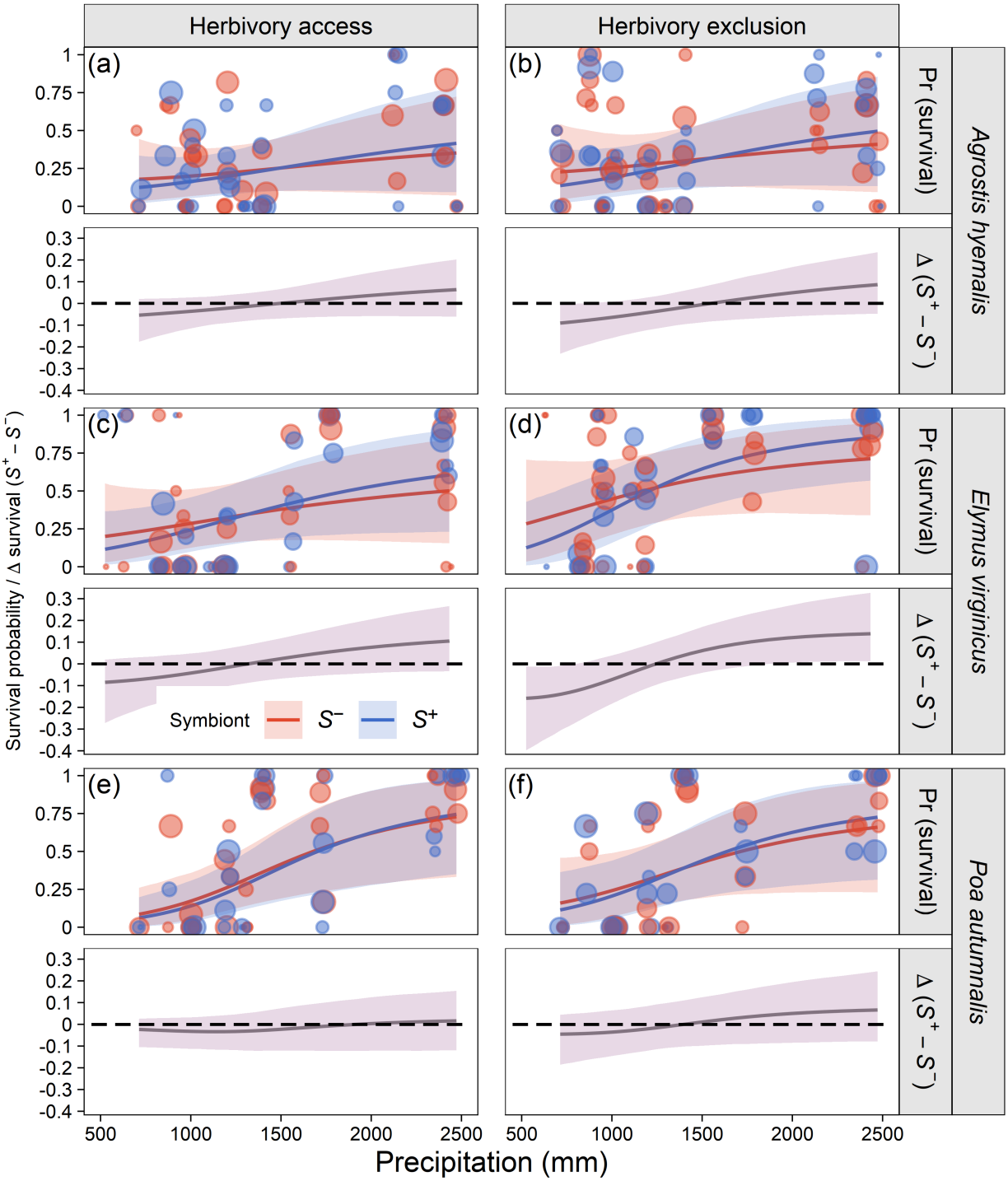


Figure 3: Predicted survival probability (upper panels) and symbiont effect ($\Delta = S^{+} - S^{-}$; lower panels) across the precipitation gradient. Lines and shaded regions show posterior means and 90% credible intervals, respectively. Points indicate observed plot means, with point size proportional to sample size. Horizontal dashed line denotes $\Delta = 0$. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

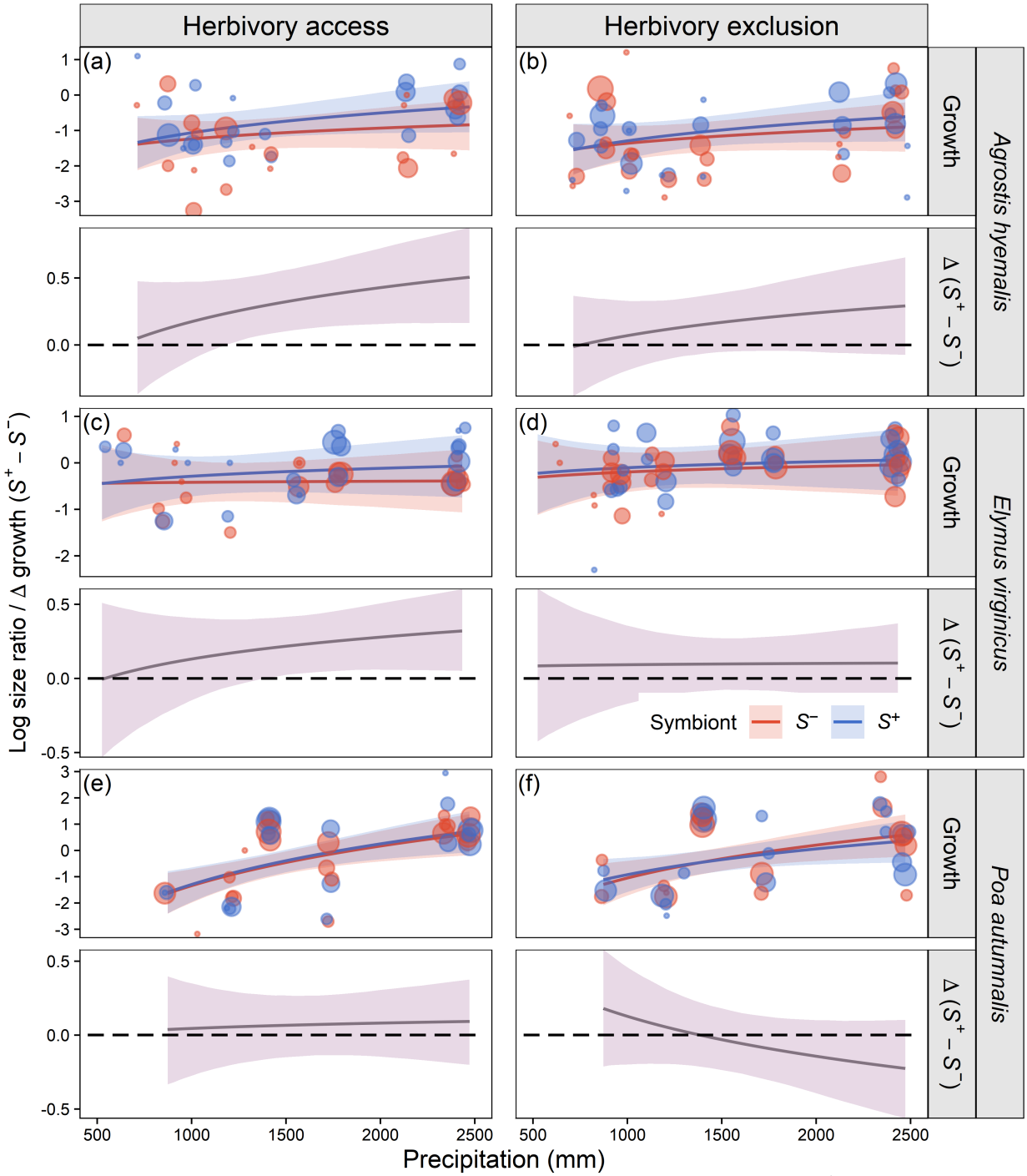


Figure 4: Predicted growth (log size ratio; upper panels) and symbiont effect ($\Delta = S^{+} - S^{-}$; lower panels) across the precipitation gradient. Lines and shaded regions show posterior means and 90% credible intervals, respectively. Points indicate observed plot means, with point size proportional to sample size. Horizontal dashed line denotes $\Delta = 0$. Curves reflect models fitted on log precipitation. Panels are arranged by species and herbivory treatment (herbivore access and exclusion).

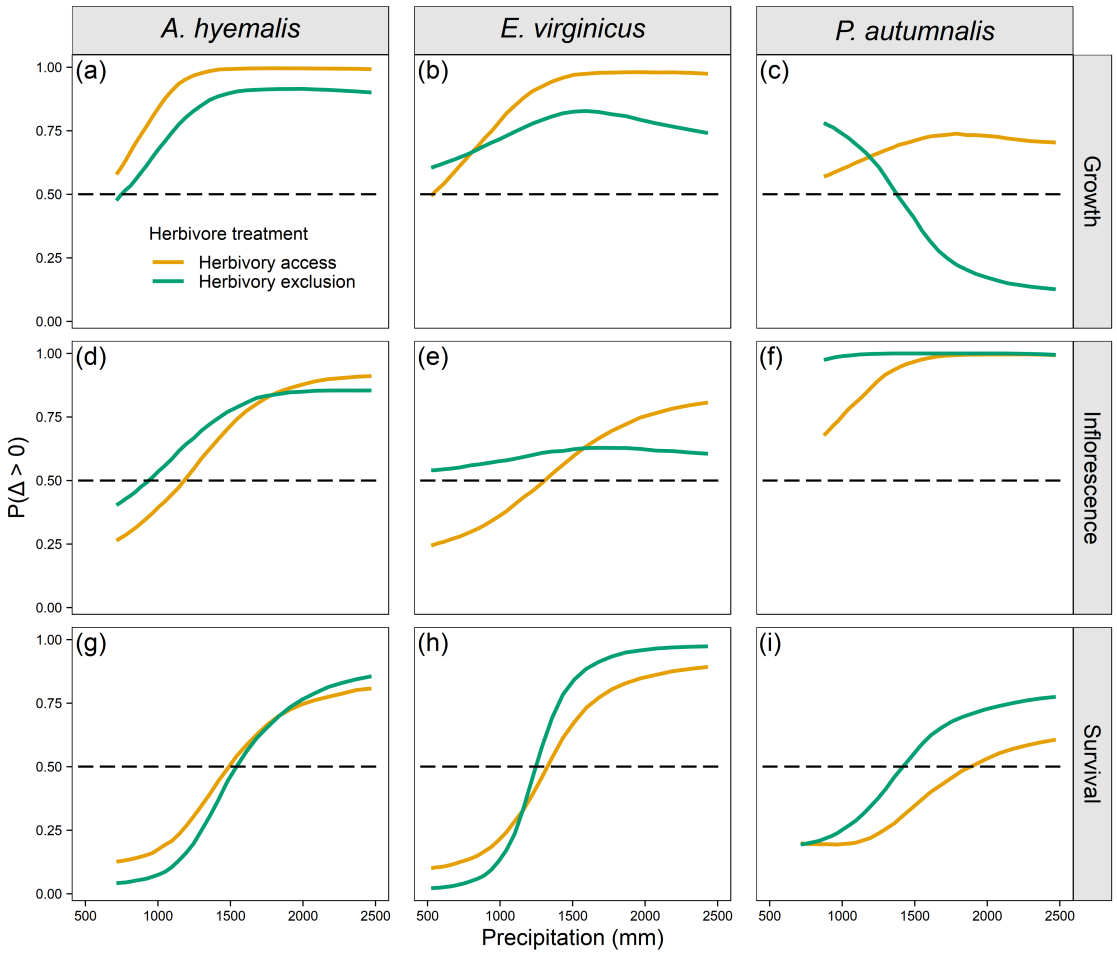


Figure 5: Probabilities of positive endophyte effects on survival, growth, and inflorescence production across precipitation gradients and under herbivore access or exclusion for *A. hyemalis*, *E. virginicus*, and *P. autumnalis*. Panels show the probability that the difference in vital rates between S^+ and S^- hosts is positive ($\Pr[\Delta > 0]$). Values closer to 1.0 indicate high confidence in an S^+ advantage, values closer to 0.0 indicate high confidence in an S^- advantage, and values close to 0.5 indicate no difference between S^+ and S^-.

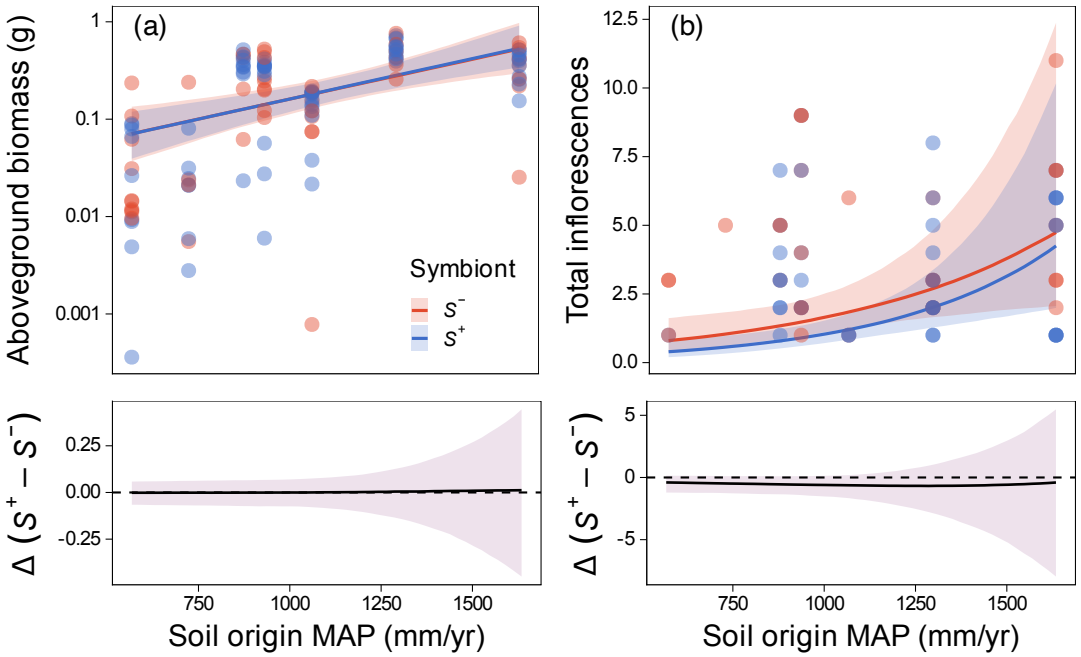


Figure 6: Greenhouse experiment testing whether soil-origin precipitation modifies symbiont effects on host performance in *Agrostis hyemalis*. Upper panels: posterior predictions for aboveground biomass (a) and total inflorescence production (b) across the 30-year mean annual precipitation of the soil-origin site (mm yr^{-1}), for symbiont-present (S^+) and symbiont-absent (S^-) plants; points are observed values, lines are posterior medians, and shaded ribbons are 95% credible intervals. Biomass was modelled on the log scale and is shown on a logarithmic axis. Lower panels: posterior contrast between symbiont treatments ($\Delta = S^+ - S^-$) for the response shown above; the dashed line marks no difference between treatments.

May 11, 2026

Dr. Peter H. Thrall
Editor-in-Chief, *Ecology Letters*
CSIRO National Collections & Marine Infrastructure
GPO Box 1600, Canberra ACT 2601, Australia

Dear Dr. Thrall,

We are pleased to submit our manuscript, *"Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses"*, for consideration as a Letter in *Ecology Letters*.

Understanding when and why mutualisms break down under environmental stress is a central challenge in ecology, with increasing urgency as climate change pushes species into novel conditions. Despite extensive theory on the stress dependence of symbiotic benefits, rigorous empirical tests remain rare, particularly those that move beyond single host species or controlled settings to evaluate symbiont effects across the full range of conditions hosts naturally encounter. Our study addresses this gap through an integrative design. We conducted common garden experiments with three grass species hosting vertically transmitted fungal endophytes (*Epichlöe* spp.) across a natural aridity gradient spanning seven sites in the south-central United States, with herbivore exclusion treatments nested within sites. We analyzed demographic vital rates using hierarchical models that partition variation across species, sites, and treatments, providing resolution unattainable in simpler designs. A complementary greenhouse experiment, which independently manipulated soil origin, allowed us to isolate mechanisms underlying field patterns. By combining multi-species field experiments, demographic modelling, and controlled greenhouse tests, our study provides a rare, mechanistic evaluation of context dependent symbiosis. We believe this approach allows us to address both generality and underlying drivers, questions of broad relevance to the readership of *Ecology Letters*.

We have opted out of double-blind review because this paper builds upon our previous work, so our identities could be easily deduced. Our identities are also evident from our GitHub repository, where we direct reviewers to find all data and code necessary to reproduce our analyses.

Sincerely yours,

Jacob K. Moutouama
On behalf of all co-authors

Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

Jacob K. Moutouama [*1, 2], Josh C. Fowler[3], Akiem Gough[4], Julia Martin[4], Chris Oxley[4], Karl Schrader[4], Karis Williams[4], and Tom E.X. Miller[4]

[1]Department of Botany, University of British Columbia, Vancouver, BC, Canada
[2]African Centre for DNA Barcoding, University of Johannesburg, Johannesburg, South Africa
[3]Department of Environmental Studies, University of Colorado Boulder, Boulder, CO, USA
[4]Department of BioSciences, Rice University, Houston, TX, USA

Running Title: Stress and Grass-Endophyte Symbioses

Keywords: demography, parasitism, plant-fungi interactions, range limits, stress gradient hypothesis, symbiosis

Article Type: Letter

Data accessibility statement: Upon acceptance, all data and computer scripts supporting the results will be archived in a Zenodo package. During peer review, all data files required to reproduce the analyses, including annual census data, are temporarily hosted via Dropbox and available at `https://www.dropbox.com/scl/fo/fgxng2mcemjyv41bihul9/AGiZs5mPRGcCZURpGyvhjCE?rlkey=7mgdyrus3ga44yj3ygfncfmkd&dl=1`. All analysis scripts (Stan and R) are available at `https://github.com/jmoutouama/endo-range-limits`.

Conflict of interest statement: None.

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Authorship statement: J.K.M. contributed to data collection, data analysis, and led manuscript drafting. J.C.F., A.G., J.M., C.O., K.S., and K.W. contributed to data collection and manuscript revisions. T.E.X.M. contributed to research conception, data collection, data analysis, and manuscript revisions.

This file contains: Abstract (150 words), Main Text (4745 words), Figures (1-6),Supporting Information - Supplemental Methods, Supplemental Figures S1-S10, References (72).

*Corresponding author: jmoutouama@gmail.com